

ROUTING DRIFT ALONE DOES NOT DIAGNOSE FAILURE IN MERGED MOE LLMs

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ABSTRACT

Model merging efficiently combines specialized large language models (LLMs) without joint retraining, but can substantially alter expert routing in Mixture-of-Experts (MoE) models. Such *routing drift* is often interpreted as routing failure, raising a fundamental question that remains unclear: *does routing drift after MoE merging actually indicate routing failure, and what evidence should justify repair?* We investigate these questions across DeepSeekMoE, OLMoE, and Qwen3-MoE proposing a routing analysis toolkit for controlled counterfactual interventions and token-level analysis. By crossing source and merged router inputs and parameters, we attribute most expert reassignments to input shifts rather than parameter changes at the same layer. However, source-relative routing differences poorly predict next-token likelihood gains from source-route restoration, and different expert selections can produce directionally similar mixture outputs. We therefore operationalize routing failure as *task loss recoverable under a specified routing intervention, with non-routing parameters fixed*. These tests detect recoverable loss under deliberate router corruption, whereas source-route restoration does not establish reliable task benefits in the evaluated merged models. Motivated by these, we propose *Selective Router Repair (SRR)* as a case study, and find that source-specialist token-likelihood advantages do not reliably identify beneficial local corrections. Together, these findings show that **routing drift alone is insufficient evidence of routing failure**: source-informed corrections must be judged by their task-level intervention effects. The analysis toolkit and SRR code are released.¹

1 INTRODUCTION

Model merging combines complementary capabilities from independently specialized large language models (LLMs) without joint retraining (Zhou et al., 2025; Yang et al., 2026; Li et al., 2026). Most existing methods target dense LLMs, where all tokens use the same feed-forward modules (Wang et al., 2026d; Zhou et al., 2026). Mixture-of-Experts (MoE) models instead route each token to only a small subset of experts (Jacobs et al., 1991; Shazeer et al., 2017; Lepikhin et al., 2020). Merging MoEs can therefore change not only model parameters but also token-to-expert routing. These changes in expert assignment and routing probabilities are referred to as *routing drift*.

Routing drift is naturally concerning: if a merged model sends a token to different experts than its corresponding specialized source model, the router may appear to have failed, motivating recent routing-realignment methods (Huang et al., 2026). However, a changed route is not necessarily a *routing failure*. We use *routing failure* to denote degradation in task-relevant behavior attributable to routing, rather than a structural change in routing itself. Whether routing drift reliably diagnoses such failure still remains unclear. This raises our central question: *does routing drift after MoE merging actually indicate routing failure, and what evidence should justify repair?*

Answering this question is nontrivial because routing drift alone reveals neither why routing changed nor whether it caused harm. A token’s route depends jointly on its router-input representation, the

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¹Our toolkit and code are available at <https://github.com/wyy-code/SRR>.

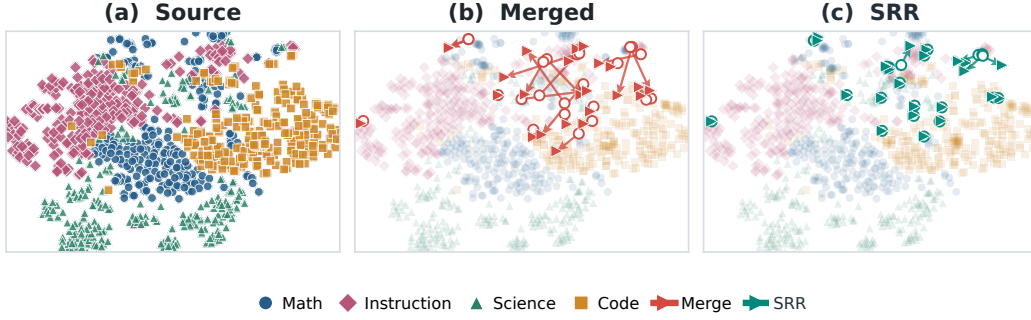


Figure 1: **Merge-induced representation drift and SRR updates.** Illustrative joint t-SNE of aligned router-input representations in OLMoE. Arrows track the same domain-balanced tokens from Source to Merged in (b) and their corresponding shifts after SRR in (c).

hidden state presented to the router, and the router parameters, so the same routing change may arise from representation shifts, router changes, or both. Moreover, expert assignment is only an intermediate structural decision: the MoE layer output also depends on the weighted outputs of the selected experts. Figure 1(a,b) illustrates post-merge representation shifts for aligned tokens. In short, routing drift therefore describes *what changed*; routing failure asks *what harm routing caused*. Thus, establishing failure requires isolating the causal effect of routing.

To address these questions, we develop a routing analysis toolkit for controlled counterfactual interventions and paired token- and task-level evaluation. Across DeepSeekMoE, OLMoE, and Qwen3-MoE under different merging methods, we cross source and merged router inputs and parameters on aligned token sequences. In 77.7–96.9% of changed-route events, input-only replacement changes the source expert set, whereas parameter-only replacement does not. This local attribution does not exclude upstream routing effects. Yet full-distribution JS divergence poorly predicts next-token likelihood gains from source-route restoration (under 0.52 AUROC). At fixed merged states and expert parameters, source and native merged routers also produce similar mixtures, exceeding matched random-route controls in cosine similarity. Task consequences therefore require a separate test.

To provide that test, we operationalize routing failure as task loss recoverable under an alternative routing policy, with non-routing parameters fixed. We compare policies on identical items using the merged model’s experts, while allowing downstream representations to respond. As a positive control, replaying clean routes after permuting OLMoE router logits recovers an accuracy loss. By contrast, source-route restoration in evaluated merged models does not establish reliable task benefits. An inconclusive result establishes neither harmlessness nor optimal routing. This grounds the diagnosis in task consequences relative to a specified intervention, not source agreement.

This criterion also lets us examine whether source predictions justify selective corrections. A source specialist’s token-likelihood advantage compares models, rather than isolating the value of a routing update. We propose *Selective Router Repair (SRR)* as a case study: it fits likelihood-weighted, source-derived expert-pair corrections on merged router inputs, updating selected router parameters. Figure 1(c) illustrates a candidate’s representation changes, not task recovery. On independent diagnostic prompts, we compare source-informed and fitted directions with native and opposite-direction controls within the merged model. These tests do not establish that positive source-likelihood advantage selects beneficial local corrections; paired evaluation on fixed checkpoints does not establish an average task gain. Together, these findings show that **routing drift alone is insufficient evidence of routing failure**: candidate construction and task recovery are distinct claims.

In summary, our contributions are summarized as follows:

- (1) **Routing drift analysis:** Across different MoE models, local interventions attribute most reassignments to input shifts, while structural differences poorly predict source-route intervention gains.
- (2) **Task-grounded diagnosis:** We designed analysis toolkit to formalize intervention-relative recoverable task loss and implement paired routing tests with fixed non-routing parameters.
- (3) **Repair supervision:** Our proposed SRR finds no reliable evidence that source-likelihood advantages identify beneficial corrections or that fitted updates improve performance on fixed checkpoints.

2 RELATED WORK

Model Merging combines specialized models without joint retraining or ensemble inference. Prior work spans parameter averaging and weighted combination (Wortsman et al., 2022; Matena & Raffel, 2022), task-vector composition (Ilharco et al., 2022), and interference reduction through sign resolution (Yadav et al., 2023), sparsification (Yu et al., 2024), adaptive coefficients (Yang et al., 2024b; Stoica et al., 2024; Wang et al., 2026a), feature alignment (Lu et al., 2024; Stoica et al., 2025; Wang et al., 2026b), and low-rank structure (Gargiulo et al., 2025; Cheng et al., 2025). Recent studies scale merging to LLMs (Akiba et al., 2025; Wang et al., 2026g), automate merge search (Wang et al., 2026c), study scaling laws (Wang et al., 2026d), and extend merging to agents and embodied models (Yuan et al., 2026; Fu et al., 2026; Li et al., 2025). These studies primarily ask *how* dense model parameters should be combined. Our focus is complementary: we investigate what routing drift mean within an already merged MoE and what evidence should justify repair.

Routing Drift and Repair. Sparse MoEs use learned routers to select token-specific experts (Fedus et al., 2022; Dua et al., 2022; Zoph et al., 2022; Zhou et al., 2022; Puigcerver et al., 2024), with modern MoE LLMs increasingly relying on large-scale expert specialization (Jiang et al., 2024; Xu et al., 2026; Liu et al., 2024; Yang et al., 2025). Routing has also guided expert merging and compression (Li et al., 2024). Most closely related, HARC treats source-to-merged routing mismatch as routing breakdown and realigns the merged router (Huang et al., 2026). Other post-merge methods instead align representations or features, including Surgery (Yang et al., 2024a), ProbSurgery (Wei et al., 2025), FeatCal (Gu et al., 2026a), and Expert Merging (Zhang et al., 2026). Our work instead asks whether routing change constitutes failure. We distinguish alignment as a repair objective from disagreement as evidence of routing failure. Our toolkit separates local input and router-parameter effects, then measures recoverable task loss under specified routing interventions with non-routing parameters fixed. Proposing SRR as a case study, we further test whether source-likelihood advantages support useful local corrections, rather than treating them as failure labels.

3 ROUTING DRIFT IS NOT ROUTING FAILURE

In this section, we develop controlled intervention analysis that disentangles representation- and router-induced changes. We examine the origin of routing drift (Sec. 3.1), whether it predicts intervention gains (Sec. 3.2), and how different routes preserve similar mixture outputs (Sec. 3.3). We then formulate and validate task-grounded tests of routing failure (Sec. 4).

3.1 MOST POST-MERGE ROUTE CHANGES ARE REPRESENTATION-INDUCED

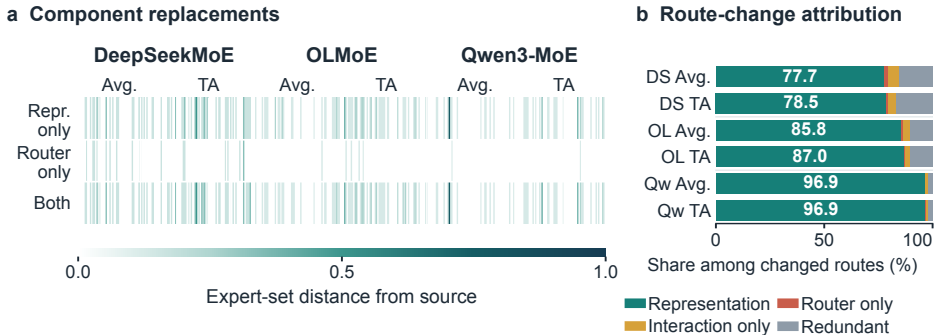


Figure 2: **Most post-merge route changes are representation-induced.** (a) Expert-set distances for 64 aligned events per setting under component replacements. (b) Full-sample attribution among changed routes. Protocols and intervals appear in Appendices A.4 and A.8.

Figure 1(a,b) illustrates shifts in router-input representations; aligned input and output views across three architectures appear in Appendix A.2. We cross source and merged representations with source and merged router parameters on identical token sequences. Across DeepSeekMoE (Dai et al., 2024), OLMoE (Muennighoff et al., 2025), and Qwen3-MoE (Yang et al., 2025), each under Average and Task Arithmetic merging, we analyze 256 domain-balanced prompts per setting, all

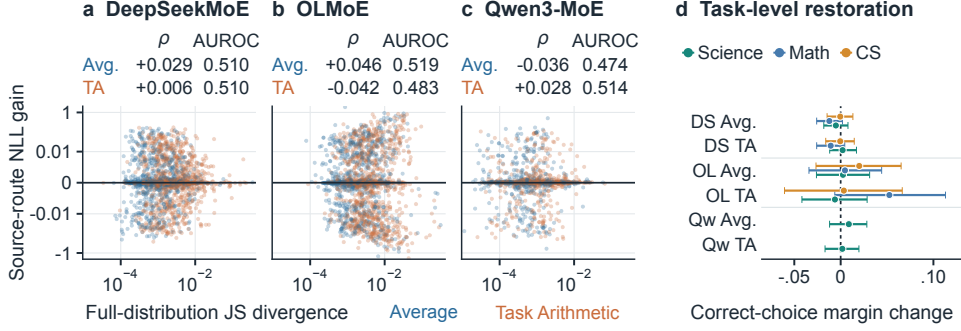


Figure 3: **Routing drift alone is insufficient evidence of routing failure.** (a–c) JS divergence versus source-route NLL gain (log- x , symmetric-log y). (d) Task-margin changes with 95% bootstrap intervals. Full tests appear in Appendices A.6–A.8.

continuation tokens, and all sparse layers. Figure 2(a) displays $1 - |S_{\text{src}} \cap S_{\text{cond}}|/k$, where the sets contain the k selected experts under source and replacement conditions. Across these six settings, 26.5–54.3% of token–layer expert sets change. Among changed routes, 77.7–96.9% change under representation-only replacement but not router-only replacement; we call these *representation-induced*. The converse, *router-parameter-induced* changes, accounts for at most 1.8% (Figure 2(b)). These categories do not require reproducing the merged expert set; full definitions and domain and details appear in Appendices A.1, A.4, and A.8. Altered router inputs thus explain most observed expert reassignments, without establishing whether they are harmful.

3.2 ROUTING DRIFT POORLY PREDICTS SOURCE-ROUTE GAINS

We test whether structural differences predict intervention gains: reductions in next-token negative log-likelihood (NLL) on fixed continuations. On changed-route events in the final five sparse layers, positive and negative source-route gains occur at overlapping full-distribution Jensen–Shannon (JS) divergences across all three architectures (Figure 3(a–c)). JS-based prediction remains near chance (AUROC 0.47–0.52; chance: 0.5). Other distances, including Top- k set distance, and intervention targets are reported in Appendices A.6 and A.8.

We then restore source-derived expert selections and native routing weights while keeping model parameters fixed. Token-local controls can increase NLL. For task evaluation, we replace routes throughout question–choice sequences and score answer tokens only. Changes in correct-choice margin—the correct option’s score advantage over the strongest alternative—have 95% confidence intervals spanning zero (Figure 3(d)), providing no reliable evidence of task improvement. Accuracy results and protocols appear in Appendices A.7 and A.8.

This does not imply optimal merged routing. Local tests on DeepSeekMoE and OLMoE show that the merged route is rarely best within fixed candidate sets, yet the source route wins only roughly half the comparisons (Appendix A.9). Thus, **routing drift alone is insufficient evidence of routing failure**, the tested structural metrics poorly predict the benefit of source-route restoration in these settings. These comparisons neither establish optimal merged routing nor exclude task loss recoverable under other routing policies.

3.3 DIFFERENT ROUTES CAN PRESERVE SIMILAR MIXTURE OUTPUTS

We compare source-route and native merged-route mixtures at the same merged hidden state h , with expert parameters fixed. At layer ℓ ,

$$o^{(\ell)}(h; S, g) = \sum_{e \in S} g_e E_e^{(\ell)}(h), \quad (1)$$

where S is the selected expert set, g_e the native weight of expert e , and $E_e^{(\ell)}(h)$ its output.

For each changed-route event, we compute the maximum output cosine between entering and leaving experts. Across the six settings, its mean is 0.041–0.096, whereas source-route and native

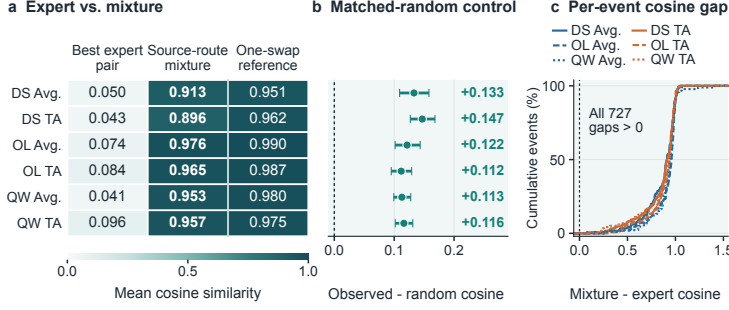


Figure 4: Different routes, similar mixture outputs. (a) Mean cosines. (b) Paired observed-minus-random cosine (95% CIs). (c) Cumulative event-wise cosine gaps.

merged-route mixtures have mean cosine 0.896–0.976 (Figure 4(a)). Mixture cosine exceeds the expert-pair maximum in all 727 sampled events (Figure 4(c)).

To control for overlap and weighting, we compare each source route with 32 random alternatives matched for expert count, overlap with the merged route, and routing-weight values. Observed routes exceed these controls in mean cosine by 0.112–0.147, with positive paired 95% prompt-bootstrap intervals in every setting (Figure 4(b); Appendix A.10). This supports *mixture-level functional redundancy*: different expert selections can preserve aligned mixture outputs, without guaranteeing unchanged magnitudes or downstream behavior.

4 TASK-GROUNDED DIAGNOSIS OF ROUTING FAILURE

4.1 INTERVENTION-RELATIVE RECOVERABLE LOSS

Section 3 separates routing changes from their local consequences. To test task harm, we compare a baseline routing policy π with a specified alternative π' within the same model. Each policy determines expert selections and weights within a fixed intervention scope. With non-routing parameters ϕ held fixed, define

$$H_{\mathcal{D}}(\pi; \pi') = \mathbb{E}_{(x,y) \sim \mathcal{D}} [\ell_{\text{task}}(M_{\phi, \pi}; x, y) - \ell_{\text{task}}(M_{\phi, \pi'}; x, y)]. \quad (2)$$

The alternative policy is part of the estimand: schedule replay, an online router swap, and a calibrated router are distinct interventions because they can induce different downstream states. Both policies are evaluated on identical task items, allowing downstream representations to respond to the intervention. For zero-one loss, $H_{\mathcal{D}}$ equals the accuracy gain; positive values indicate recoverable loss relative to the chosen task, alternative, and scope. A confidence interval wholly above zero supports this recovery; an interval containing zero establishes neither harmlessness nor optimality. Recovery through routing does not by itself identify router-parameter changes as the original cause of degradation. Native-routing checks, scoring, and uncertainty procedures appear in Appendix C.

4.2 CONTROLLED RECOVERY AND NATURAL ROUTING ALTERNATIVES

Known routing perturbations. We permute expert logits in nested sets of $k \in \{1, 4, 16\}$ OLMoE layers under two fixed randomizations per Average and Task Arithmetic parent. On 256 fixed ARC items per parent, we compare corrupted execution with replay of that parent’s clean route schedule. Figure 5(a) shows the accuracy changes; panel (b) retains all twelve paired effects and their uncertainty. Four-layer perturbations yield 12.50–14.84 percentage points (pp) of recovery, and full-layer perturbations yield 26.56–32.81 pp. All eight comparisons at $k = 4, 16$ pass Holm correction across the twelve tests; none at $k = 1$ does. One-layer perturbations do not consistently lower accuracy. Replay accuracy ranges from 50.39%

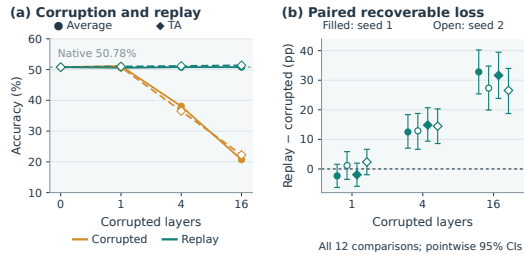


Figure 5: Task effects of controlled routing interventions. (a) Corrupted vs. clean-route replay accuracy. (b) Paired replay gains with 95% bootstrap CIs across all parent-permutation settings.

to 51.56% around the 50.78% clean reference, so this is not reconstruction of the clean forward pass. These controls establish recovery of the tested imposed losses, not a repair learned by SRR.

Natural routing alternatives. A natural reference need not be appropriate for the merged experts. Table 1 brings together the science-domain source replays from Section 3.2 and separate OLMoE replays of a frozen LC-calibrated route schedule. This intervention replays source-derived routing decisions computed from the source forward; it is not a source-router parameter swap. A swapped source router would instead recompute routing on hidden states produced by the intervened merged network, and therefore defines a different counterfactual policy. Neither accuracy nor correct-choice margin—the correct option’s score minus the highest incorrect-option score—has a strictly positive interval in any displayed setting. LC replay changes correctness on just one item per parent, yielding -0.39 pp for Average and $+0.39$ pp for TA. These results do not establish recovery under the tested natural references. Thus, routing failure requires evidence of recoverable task loss under a specified intervention, not source disagreement alone. We next use source information to construct selective router updates and evaluate their effects separately from their fitting objective.

5 SELECTIVE ROUTER REPAIR

Section 4 separates candidate interventions from evidence of recovery. We construct **Selective Router Repair (SRR)** to study selective source supervision, then test its local directions, executed corrections, and task effects.

5.1 SELECTING EXPERT PAIRS

Let B , S_d , and M denote the shared base, domain- d specialist, and merged parent. On identical parent-generated continuations, each model uses its own hidden states. For response token y_t after prefix c_t , write $n_t^A = -\log P_A(y_t | c_t)$ and let $z_t^A \in \mathbb{R}^E$ be the router logits over E experts. Omitting layer indices and writing $S = S_d$, we form a source–base preference profile for each prompt i :

$$\bar{u}_i = \frac{\sum_{t \in i} \omega_t \text{center}(z_t^S - z_t^B)}{\sum_{t \in i} \omega_t}, \quad \omega_t = \sigma((n_t^B - n_t^S)/\tau), \quad (3)$$

where σ is the sigmoid and center subtracts the expert-wise mean. Activity is the similarly weighted maximum of source and base routing probabilities. We average both quantities equally over prompts in each domain and construction split. Pairs join active experts with opposite, consistent profile signs across both splits. Ranking combines profile separation, sign consistency, and activity; selection cycles across domains without reusing an expert. The fixed screen requires activity ≥ 0.002 and sign consistency ≥ 0.75 , selecting two pairs per domain and layer without benchmark scores.

5.2 FITTING ROUTER CORRECTIONS

For pair $j = (a_j, b_j)$ assigned to domain d_j , let $m_{tj}^A = z_{t,a_j}^A - z_{t,b_j}^A$ and $r_{tj} = m_{tj}^S - m_{tj}^M$. We clip this source–parent residual by the source–base difference and weight matching-domain tokens on which the specialist outpredicts the parent:

$$\begin{aligned} \tilde{r}_{tj} &= \text{sgn}(r_{tj}) \min\{|r_{tj}|, |m_{tj}^S - m_{tj}^B|\}, \\ \rho_{tj} &= \mathbf{1}[d_t = d_j] \mathbf{1}[\delta_t > 0] \sigma(\delta_t/\tau), \quad \delta_t = n_t^M - n_t^S. \end{aligned} \quad (4)$$

Unlike ω_t , which measures source advantage over the base, ρ_{tj} compares the source with the parent. Neither weight is a label of routing failure. For cached parent inputs $h_t^M \in \mathbb{R}^{d_h}$ and N training

Table 1: **Natural routing alternatives.** Changes from native routing with paired 95% bootstrap CIs. m denotes the correct-choice margin.

Model / merge	N	ΔAcc (pp)	Δm (10^{-3})
Source-route replay: science			
DeepSeekMoE / Avg.	256	0.00 [−1.56, +1.56]	−5.2 [−18.2, +7.9]
DeepSeekMoE / TA	256	+0.78 [−0.78, +2.34]	+2.1 [−11.8, +17.0]
OLMoE / Avg.	256	+1.17 [−1.56, +3.91]	+2.5 [−26.1, +31.2]
OLMoE / TA	256	+1.17 [−1.17, +3.52]	−6.2 [−41.8, +28.5]
Qwen3-MoE / Avg.	64	−1.56 [−4.69, 0.00]	+8.8 [−12.0, +28.2]
Qwen3-MoE / TA	64	−1.56 [−4.69, 0.00]	+1.8 [−16.9, +19.7]
LC replay: ARC			
OLMoE / Avg.	256	−0.39 [−1.17, 0.00]	−0.3 [−3.7, +3.2]
OLMoE / TA	256	+0.39 [0.00, +1.17]	−2.6 [−6.6, +0.8]

tokens, we solve

$$v_j^* = \arg \min_v \frac{1}{2} \sum_{t=1}^N \rho_{tj} ((h_t^M)^\top v - \tilde{r}_{tj})^2 + \frac{\lambda_j}{2} \|v\|_2^2, \quad (5)$$

$$\lambda_j = \gamma \max \left\{ N^{-1} \sum_{t=1}^N \rho_{tj} \|h_t^M\|_2^2, 10^{-12} \right\}.$$

Writing $w_e^\top$ for router row e , the update is

$$w'_{a_j} = w_{a_j}^M + \frac{\eta}{2} v_j^*, \quad w'_{b_j} = w_{b_j}^M - \frac{\eta}{2} v_j^*. \quad (6)$$

This changes the pair gap by $\eta h^\top v_j^*$ on fixed input h . Only selected rows in the final five sparse layers change; all fits precede joint writeback, without refreshing traces. We use $\tau = 0.5$, $\gamma = 10^{-3}$, $\eta = 0.0625$, and preconditioned conjugate gradients (Appendix D). The objective fits selected logit residuals rather than task loss. A two-expert counterexample shows why better proxy fit need not improve token likelihood (Appendix D.4).

5.3 LOCAL SUPERVISION AND EXECUTED CORRECTIONS

Do source predictions identify useful directions? We evaluate 256 independent diagnostic prompts for each OLMoE and Qwen3-MoE under Average and TA candidate, selecting one token–layer–pair event per prompt. At event e , equal-magnitude positive and opposite logit perturbations follow either the source-informed or fitted direction q :

$$U_e^q = \ell_e^{\text{native}} - \ell_e^{q,+}, D_e^q = \ell_e^{q,-} - \ell_e^{q,+}, \quad (7)$$

where ℓ_e is next-token NLL. Positive U improves over native; positive D favors the proposed direction over its opposite. Of 1024 events, 471 have positive source advantage. Source and fitted directions agree in sign on 83.4% of supported events (95% CI [80.0, 86.6]%), yet neither pooled U nor D establishes an improvement (Figure 6(a,b), OL and QW denote OLMoE and Qwen3-MoE). The matched source-direction comparison also does not establish enrichment (Appendix E). Directional agreement therefore does not supply the missing evidence of local utility.

Do fitted corrections execute as predicted? All fits use parent inputs, but earlier updates can change later inputs. On identical continuations through eight OLMoE/Qwen3-MoE Parent/SRR pairs, we measure one event per prompt in each updated layer: 128 prompts per setting and 5,120 events in total. At reference precision, the selected pair’s logit change decomposes as

$$\Delta m_{t,ab} = \underbrace{(\Delta w_a - \Delta w_b)^\top h_t^M}_{d_{t,ab}: \text{direct correction}} + \underbrace{(w'_a - w'_b)^\top (h_t^{\text{SRR}} - h_t^M)}_{r_{t,ab}: \text{input response}}. \quad (8)$$

The input response is zero at the first updated layer. At the final layer, its reported RMS is approximately 0.38–0.86 times the direct-correction RMS across settings (Figure 6(c)). Thus, joint execution need not reproduce the fixed-input correction. The figure summarizes the reference-precision decomposition; native-forward discrepancies, absolute RMS intervals, and routing-set changes are reported in Appendix F. This decomposition does not attribute task effects to input response.

6 EXPERIMENTS

Section 5 examines the construction and execution of SRR. We now test whether these updates recover downstream performance under Section 4’s task-grounded criterion.

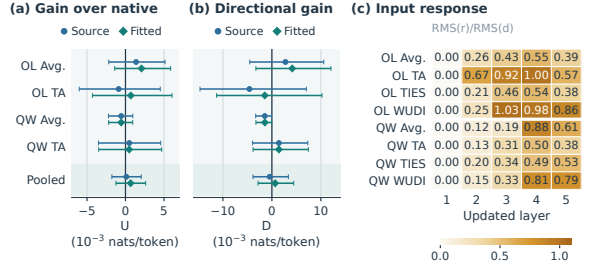


Figure 6: **Local utility and input response of SRR.** (a,b) Positive-support NLL gains over native (U) and opposite (D) routing, with 95% bootstrap CIs. (c) Input-response/direct RMS ratios across eight settings and five updated layers.

Table 2: **Task scores across three merged MoEs.** Avg. equally weights eight benchmarks; Δ Avg. compares each method with its own Parent (pp). The 95% CI (pp) is the paired item-bootstrap 95% confidence interval for the SRR–Parent difference in percentage points. HARC scores use the same matched requests.

Method	Benchmarks (%)								Summary		95% CI (pp)	
	Hella MMLU	Swag	ARC-C	ARC-E	PIQA	Wino Grande	BoolQ	GSM8K	Avg.	Δ Avg.	Lower	Upper
DeepSeekMoE-16B-A3B												
Average	57.60	80.02	57.68	83.50	82.48	76.09	81.74	48.67	70.97	—	—	—
+ HARC	57.58	80.05	57.68	84.01	82.32	76.48	82.05	49.51	71.21	+0.236	—	—
+ SRR	57.68	79.99	57.68	83.50	82.48	76.09	81.80	48.67	70.99	+0.015	-0.011	+0.041
Task Arithmetic	58.04	80.03	58.11	83.92	82.32	76.48	80.83	45.26	70.62	—	—	—
+ HARC	58.07	80.05	57.94	84.01	81.94	76.72	80.61	45.56	70.61	-0.012	—	—
+ SRR	58.04	80.03	58.11	83.92	82.32	76.48	80.83	45.26	70.62	+0.000	+0.000	+0.000
TIES	56.50	80.34	57.76	82.03	82.26	76.01	82.72	54.81	71.56	—	—	—
+ HARC	56.42	80.46	56.91	82.58	82.43	76.48	82.51	53.60	71.42	-0.133	—	—
+ SRR	56.43	80.33	57.76	82.03	82.15	76.01	82.72	54.81	71.53	-0.024	-0.054	+0.004
WUDI-Merge	56.88	79.47	55.12	81.36	82.15	77.03	82.29	52.39	70.84	—	—	—
+ HARC	56.78	79.41	55.12	81.23	82.26	77.11	82.17	52.92	70.87	+0.039	—	—
+ SRR	56.97	79.51	55.17	81.43	82.21	77.03	82.32	52.51	70.89	+0.057	-0.025	+0.059
OLMoE-7B-A1B												
Average	55.47	80.15	57.08	81.52	81.83	73.72	80.40	58.15	71.04	—	—	—
+ HARC	55.56	80.18	57.25	81.44	81.61	74.11	80.52	58.98	71.21	+0.168	—	—
+ SRR	55.47	80.15	57.17	81.44	81.83	73.09	80.40	59.89	71.18	+0.139	-0.063	+0.336
Task Arithmetic	55.45	79.78	57.59	82.32	81.34	72.93	79.36	56.79	70.69	—	—	—
+ HARC	55.41	79.84	57.68	82.37	81.39	72.14	79.57	57.54	70.74	+0.048	—	—
+ SRR	55.62	79.93	57.52	82.42	81.38	73.11	79.43	57.11	70.82	+0.121	-0.163	+0.208
TIES	54.37	80.19	56.14	80.01	81.34	71.82	81.07	55.50	70.06	—	—	—
+ HARC	54.56	80.13	56.66	79.80	81.39	71.59	81.28	55.42	70.10	+0.048	—	—
+ SRR	54.49	80.29	55.99	79.98	81.49	72.24	81.14	55.60	70.15	+0.096	-0.230	+0.222
WUDI-Merge	54.12	79.52	52.39	76.26	81.23	72.53	81.99	56.10	69.27	—	—	—
+ HARC	54.19	79.56	52.73	76.39	80.96	72.53	81.62	55.57	69.19	-0.075	—	—
+ SRR	54.13	79.51	52.47	76.35	81.23	72.85	82.02	55.42	69.25	-0.021	-0.233	+0.188
Qwen3-30B-A3B												
Average	81.96	83.14	66.72	89.18	82.43	75.93	87.46	90.30	82.14	—	—	—
+ HARC	81.90	83.01	67.32	89.18	82.05	76.24	87.52	90.37	82.20	+0.060	—	—
+ SRR	81.88	83.13	66.47	89.10	82.32	76.40	87.37	90.14	82.10	-0.039	-0.212	+0.130
Task Arithmetic	81.77	83.19	67.32	89.35	83.08	76.87	88.10	87.72	82.18	—	—	—
+ HARC	81.80	83.13	67.58	89.52	82.81	77.66	87.86	87.72	82.26	+0.084	—	—
+ SRR	81.70	83.11	67.32	89.44	82.86	76.87	88.13	88.10	82.19	+0.016	-0.155	+0.184
TIES	80.95	82.50	67.24	88.34	82.26	75.53	87.86	89.84	81.82	—	—	—
+ HARC	81.58	82.61	67.75	88.01	82.21	75.06	87.37	88.86	81.68	-0.136	—	—
+ SRR	81.05	82.45	67.49	88.09	82.32	75.53	87.86	89.84	81.83	+0.013	-0.174	+0.194
WUDI-Merge	81.83	83.49	62.54	86.03	81.99	74.51	87.09	87.79	80.66	—	—	—
+ HARC	81.76	83.53	62.80	85.86	81.72	74.98	87.00	87.49	80.64	-0.016	—	—
+ SRR	81.91	83.52	62.71	85.90	81.61	74.51	87.09	87.57	80.60	-0.056	-0.261	+0.150

6.1 EXPERIMENTAL SETUP

Baselines: We study DeepSeekMoE-16B-A3B (Dai et al., 2024), OLMoE-7B-A1B (Muennighoff et al., 2025), and Qwen3-30B-A3B (Yang et al., 2025), compared with HARC (Huang et al., 2026).

Merging Settings: For each architecture, we evaluate Average merge (Ilharco et al., 2022), Task Arithmetic (TA) (Ilharco et al., 2022), TIES (Yadav et al., 2023), and state-of-the-art WUDI-Merge (Cheng et al., 2025). These methods construct the merged parents; SRR is then applied to each parent. Following Sec. 5, it fits corrections on fixed parent-generated continuations and updates selected router parameters in the final five sparse layers, leaving all other parameters unchanged.

Benchmarks and uncertainty. We evaluate MMLU (Hendrycks et al., 2020), HellaSwag (Zellers et al., 2019), ARC-Challenge (ARC-C), ARC-Easy (ARC-E) (Clark et al., 2018), PIQA (Bisk et al., 2020), WinoGrande (Sakaguchi et al., 2021), BoolQ (Clark et al., 2019), and GSM8K (Cobbe et al., 2021) with five-shot lm-evaluation-harness and a VLLM backend. Each pair shares 35,326 item identities and is evaluated five times under the fixed deterministic protocol. We retain each task’s accuracy, normalized-accuracy, or exact-match scoring rule. Avg. is the equally weighted mean of the eight benchmark percentages; $\Delta\text{Avg.}$ is SRR minus its own parent, in percentage points (pp). Paired item-bootstrap intervals resample matched records within tasks and recompute the task aggregate. They quantify item uncertainty conditional on fixed checkpoints, not variation across repair seeds. Full scoring and checkpoint details appear in Appendix G.

Evaluation Settings: All performance scores are averaged over five evaluation runs. Scores are percentages, and Avg. is their mean score. We report $\Delta\text{Avg.}$ relative to the corresponding unrepaired parent in percentage points (pp), and average these changes equally across the 12 settings for the overall result. Base-model scores provide context but are not the repair baseline.

Implementations: All experiments are conducted on 8 A800 GPUs with same configuration and software environment to ensure consistency and reproducibility.

6.2 MAIN RESULTS

Average effects across settings. Table 2 compares HARC and SRR with their corresponding merged parents. Among the eight OLMoE/Qwen3-MoE settings, five average-score changes are positive, ranging overall from -0.056 to $+0.139$ percentage points (pp). OLMoE has positive point estimates under Average, TA, and TIES, whereas Qwen3-MoE has positive estimates under TA and TIES. WUDI-Merge yields negative estimates in both.

Comparison with HARC. Across the eight active SRR settings, each method has the higher point estimate in four cases. HARC leads under Average merging and SRR under TIES on both architectures, while the ordering under TA and WUDI depends on the backbone. Relative ranking need not imply recovery: on OLMoE-WUDI, SRR exceeds HARC (-0.021 versus -0.075 pp), yet both fall below Parent. On DeepSeekMoE, HARC likewise ranges from -0.133 pp under TIES to $+0.236$ pp under Average merging.

Task-level heterogeneity. Similar averages can conceal different task responses. On OLMoE-Average, SRR gains 1.74 pp on GSM8K but loses 0.63 pp on WinoGrande; HARC gains 0.83 and 0.39 pp, respectively. On Qwen3-TA, SRR improves GSM8K by 0.38 pp but decreases PIQA by 0.22 pp. These differences do not establish task-level significance. The cross-backbone comparison also changes architecture and parameter count together, so it does not isolate a model-size effect.

Incremental value of likelihood selection. Task scores evaluate complete updates, not the contribution of their selection rule. We therefore examine the independent local diagnostic from Section 5.3. Positive-support events are matched with replacement to non-positive controls within the same architecture, merge, domain, layer, and expert pair, choosing the nearest native NLL. Across 440

Table 3: **Matched selection effects.** Positive-minus-control source-direction gains (10^{-3} nats/token; 440 matched pairs).

Contrast	Estimate	95% CI
Native (U)	-0.341	$[-3.323, 2.631]$
Opposite (D)	-0.997	$[-5.653, 3.868]$

matched pairs, neither source-direction contrast establishes higher utility on positive support (Table 3). Unlike Figure 6’s positive-support means, these differences test whether selection adds useful local information. They do not establish that the groups are equivalent or explain the benchmark changes; matching and additional weighting diagnostics appear in Appendix E.2.

Implications for routing diagnosis. These fitted updates differ from the source-route replay tested in Section 3.2. Positive point estimates therefore do not contradict the finding that source disagreement alone is insufficient evidence of failure. Following Section 4, recovery must be established for the specific intervention and task; neither fitting source-derived targets nor outperforming another repair method substitutes for evidence of improvement over the unrepaired parent.

7 CONCLUSION

In this paper, we show that **routing drift alone is insufficient evidence of routing failure** in merged MoE LLMs. Across DeepSeekMoE, OLMoE, and Qwen3-MoE, local interventions attribute most reassignments to router-input shifts, while structural differences poorly predict source-route intervention gains. Motivated by this, we formalize intervention-relative recoverable task loss and develop an *analysis toolkit* for paired routing tests with fixed non-routing parameters. These tests detect recoverable loss under controlled router corruption. To examine source-informed supervision, we propose *Selective Router Repair (SRR)*, which fits likelihood-weighted expert-pair corrections on merged router inputs. Its evaluation finds no reliable evidence that source-likelihood advantages identify beneficial local corrections or that fitted updates improve average task performance on the tested checkpoints. Together, these contributions separate candidate construction from demonstrated recovery: task-level intervention effects, rather than source agreement, must justify repair.

AI USE STATEMENT

We used generative AI tools to polish the manuscript and figures, help test experiments, and check code and infrastructure. The authors reviewed the AI-assisted work, verified the reported results and claims, and take full responsibility for the submission.

REPRODUCIBILITY STATEMENT

The anonymized code, toolkit, and configurations are available at <https://anonymous.4open.science/r/SRR-DDD2>. Section 6.1 specifies the training settings and benchmarks; Appendix G document the configuration and validation procedure.

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A ROUTING ANALYSIS: PROTOCOLS AND COMPLETE RESULTS

This appendix supports Section 3. We specify the sampling and intervention conventions, then report the complete structural, local-likelihood, and mixture-output comparisons. These measurements use different event populations and are not pooled with the SRR task evaluation.

A.1 CROSSED INTERVENTIONS

Let $\mathcal{R}(W, h) = \text{Top-}k(Wh)$ denote the native expert selection, with router parameters W and input h . For source and merged quantities, the four states are

$$\begin{aligned} r_{SS} &= \mathcal{R}(W_S, h_S), & r_{MS} &= \mathcal{R}(W_M, h_S), \\ r_{SM} &= \mathcal{R}(W_S, h_M), & r_{MM} &= \mathcal{R}(W_M, h_M). \end{aligned} \quad (9)$$

The first subscript identifies the router and the second its input. Conditioning on $r_{SS} \neq r_{MM}$, we call a change representation-induced when $r_{SM} \neq r_{SS}$ but $r_{MS} = r_{SS}$; router-parameter-induced is the converse. Both crossed routes changing the source set defines a redundant case; neither changing it defines an interaction-only case. These categories concern whether an isolated replacement changes the source set, not whether it reproduces the merged set. They establish local sufficiency under the crossed intervention, not the origin of every upstream representation change.

A.2 ALIGNED REPRESENTATIONS UNDER MERGING AND SRR

Figures A1–A3 use joint t-SNE to visualize aligned token representations from the source specialist, merged parent, and SRR checkpoint. Each figure shows router inputs above and layer outputs below: routed-mixture outputs for OLMoE, and MoE-block outputs for DeepSeekMoE and Qwen3-MoE. Colors and shapes identify domains, not experts. Red and teal arrows highlight token displacements associated with merging and SRR, respectively.

OLMoE. At layer 15, domain-associated regions in the router-input projection contrast with interleaved domain colors in the routed-mixture view (Figure A1). Merge and SRR arrows show token-dependent changes at both measurement sites, extending the input-only illustration in Figure 1.

DeepSeekMoE. At layer 27, the router-input and MoE-block-output views both contain heterogeneous token shifts (Figure A2). The block-output view retains visible domain overlap while highlighting token-specific changes under SRR.

Qwen3-MoE. The layer-47 projections contain overlapping domain regions (Figure A3). In the block-output view, the highlighted SRR paths follow different directions, illustrating heterogeneous changes within the projected layout.

Relation to the quantitative analyses. These views complement the input-shift analysis in Section 3.1 and the execution analysis in Section 5.3. They compare model representations, unlike Section 3.3’s route comparisons at a fixed merged state with fixed experts. Projected arrow lengths do not quantify original-space changes, and are not compared across rows or architectures. Task recovery is evaluated separately under Section 4’s criterion.

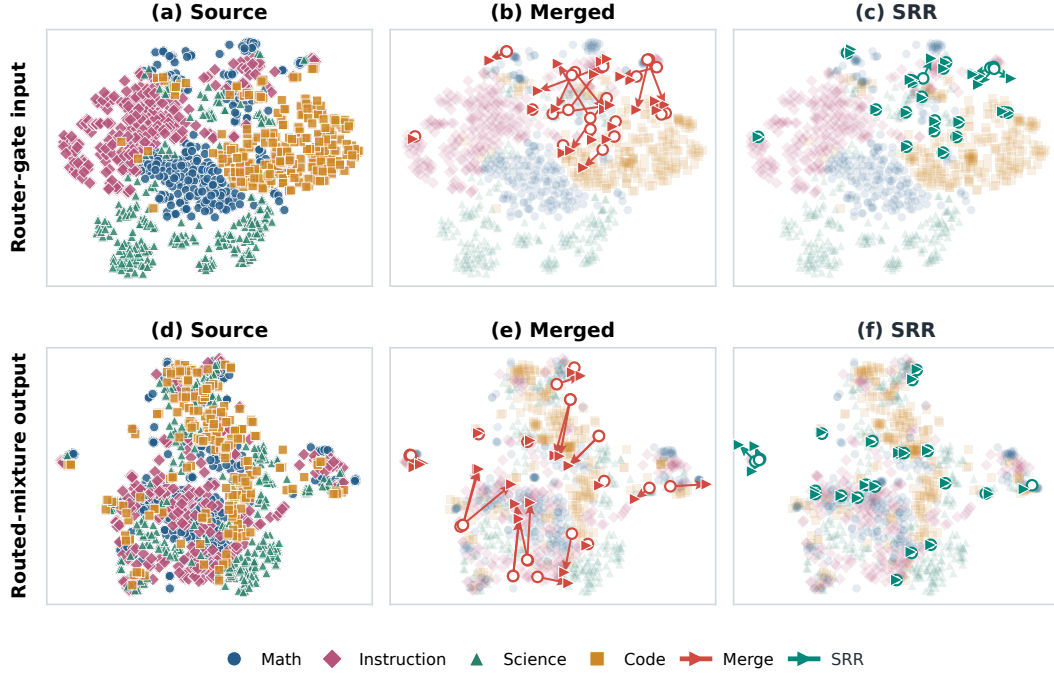


Figure A1: **Aligned representations in OLMoE.** Joint t-SNE at layer 15: router inputs (top) and routed-mixture outputs (bottom) under Source, Merged, and SRR. Red and teal arrows highlight displayed token displacements associated with merging and SRR, respectively.

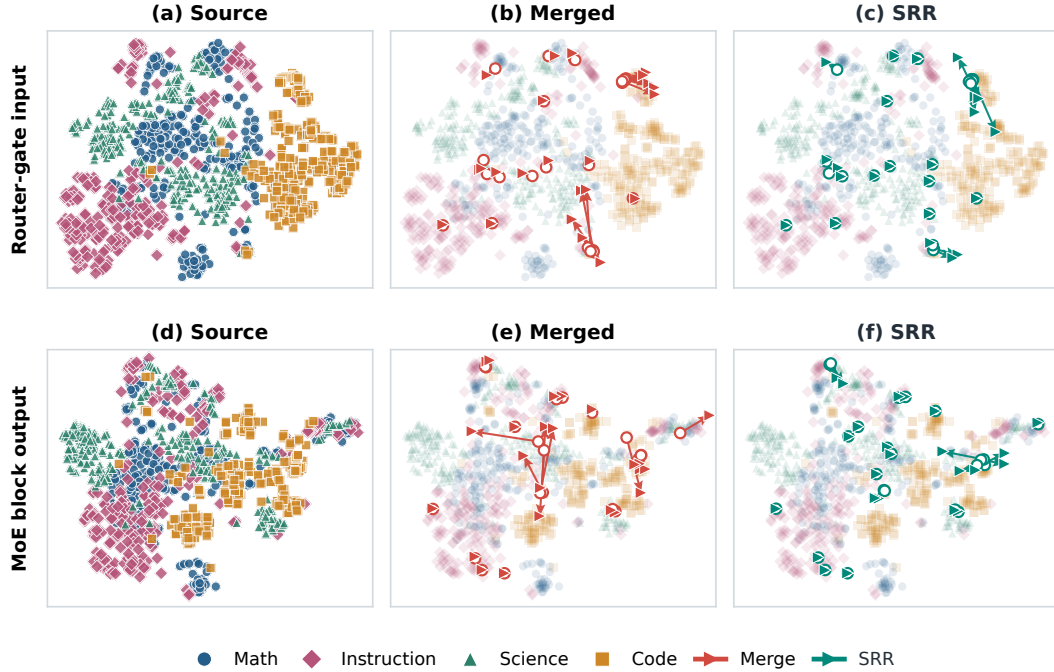


Figure A2: **Aligned representations in DeepSeekMoE.** Joint t-SNE of router inputs (top) and MoE-block outputs (bottom) at layer 27 under Source, Merged, and SRR. Display conventions follow Figure A1.

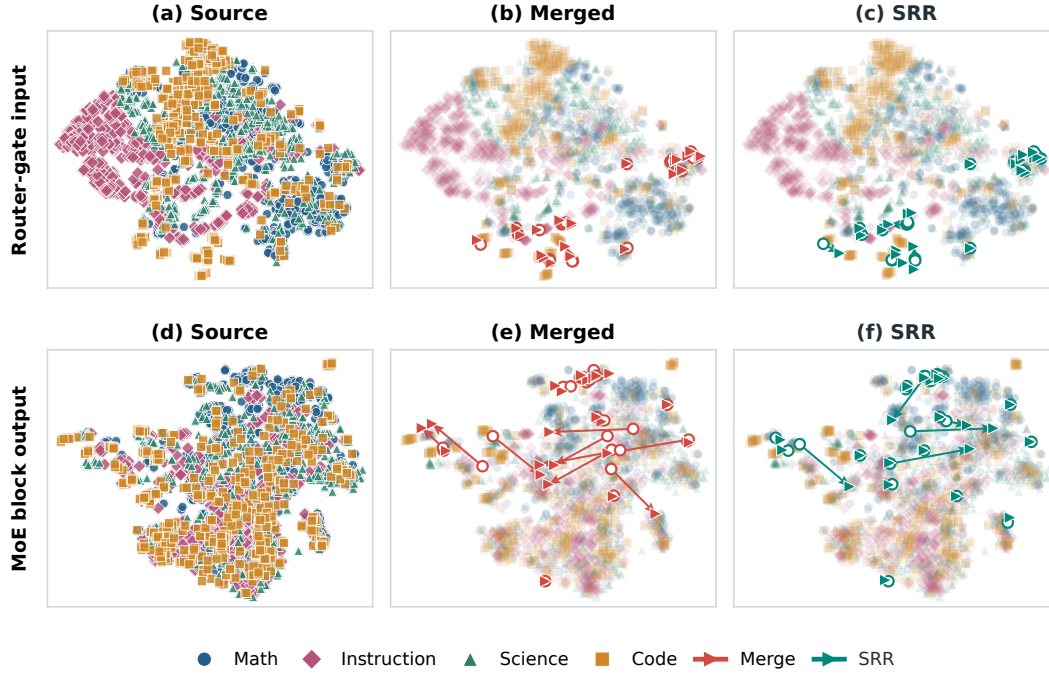


Figure A3: **Aligned representations in Qwen3-MoE.** Joint t-SNE of router inputs (top) and MoE-block outputs (bottom) at layer 47 under Source, Merged, and SRR. Display conventions follow Figure A1.

A.3 SAMPLING AND UNCERTAINTY

Table A1 records the original DeepSeekMoE/OLMoE sampling frames. Qwen3-specific counts are given in Appendix A.8. Prompts are balanced across mathematics, instruction following, science, and code; hash selections do not inspect intervention outcomes. Natural-frequency attribution includes all continuation tokens and sparse layers, whereas local functional tests concentrate on the final five sparse layers. The original natural-frequency records contain 861,651 DeepSeekMoE and 510,656 OLMoE token-layer events.

Uncertainty resamples prompts, retaining their dependent token-layer records. Attribution uses 10,000 draws and recomputes record-weighted ratios; metric associations use 4,000 draws with ranks recomputed each time. Mixture comparisons first average within prompts and use 10,000 prompt resamples. Task comparisons instead resample matched items 10,000 times. Intervals are percentile intervals unless a separate multiplicity procedure is specified. These choices preserve each estimand rather than treating all tokens or layers as independent observations.

A.4 ROUTING ORIGIN ACROSS ARCHITECTURES AND DOMAINS

DeepSeekMoE uses Top-6 routing over 27 sparse layers; OLMoE uses Top-8 over 16. Table A2 reports record-weighted attribution under Average and Task Arithmetic (TA). The corresponding Qwen3 Top-8, 48-layer results are in Table A7. Attribution conditions on observed set changes, so these shares are distinct from the prevalence of changes.

Figures A4 and A5 retain the full depth and domain profiles for DeepSeekMoE/OLMoE. Representation-induced changes dominate throughout OLMoE and most DeepSeekMoE layers. Later DeepSeekMoE layers have more redundant cases, but few router-parameter-only cases. Across domains, representation-induced shares range from 75.1% to 90.2%, versus 0.1% to 2.6% for router-parameter-induced changes. The two domain heatmaps use separate, explicitly labeled color scales.

Table A1: Sampling frames for the original DeepSeekMoE/OLMoE analyses. Hash selections do not inspect outcomes. Qwen3-specific counts appear in Appendix A.8.

Analysis	Sampling frame	Record / inference unit
Natural route origin	256 prompts (64/domain); all continuation tokens and sparse layers	Token-layer event / prompt UUID
Structural diagnostic	Same 256 prompts; final five sparse layers; eight hash-selected events per prompt (2,048/cell)	Counterfactual event / prompt UUID
Token-local route patch	80 changed-route events per architecture-parent cell	Counterfactual event / prompt UUID
Local reference and redundancy	One hash-selected event per prompt (256/cell); 112–149 changed-route events/cell	Prompt event / prompt UUID
Task-grounded route patch	256 fixed items/domain/cell; every non-padding question-choice token and sparse layer	Multiple-choice item / item
Matched random routes	32 matched alternatives for each changed-route event	Prompt event / prompt UUID

Table A2: Local four-state attribution. Values are percentages with 95% prompt-bootstrap intervals; attribution shares condition on a changed source-merged expert set.

Architecture / Parent	Changed route (%)	Representation (%)	Router parameters (%)
DeepSeekMoE / Average	33.9 [32.2, 35.7]	77.7 [77.1, 78.2]	1.79 [1.65, 1.94]
DeepSeekMoE / Task Arithmetic	46.7 [44.9, 48.7]	78.5 [78.1, 79.0]	1.14 [1.04, 1.23]
OLMoE / Average	42.7 [41.3, 44.1]	85.8 [85.4, 86.1]	0.62 [0.57, 0.68]
OLMoE / Task Arithmetic	54.3 [52.9, 55.7]	87.0 [86.6, 87.4]	0.36 [0.33, 0.40]

A.5 ACROSS-LAYER REPRESENTATION ASSOCIATION

Across 16 OLMoE layers, source-relative hidden-state cosine distance and Top- k disagreement are positively associated (Pearson $r = 0.730$, $p = 0.0013$; Spearman $\rho = 0.653$, $p = 0.0061$; Figure A6). Both vary with depth; this association does not replace the crossed intervention.

A.6 STRUCTURAL PREDICTORS AND MULTIPLICITY

The eight predictors are set distance, selected probability-mass change, ℓ_1 and JS distances between selected weights on their union, full-distribution JS and symmetric KL, entropy change, and mean absolute rank change. We compare them with source-policy, source-hidden-state, source-router, and set-only NLL gains on all events and changed-set events: 256 tests across the four DeepSeekMoE/OLMoE settings.

The frozen primary family comprises 16 changed-set tests: set distance and full-distribution JS against source-policy and source-hidden-state gains. Holm correction is applied separately to Spearman and AUROC within each declared family. None of these primary tests survives correction (Table A3); adjusted primary p -values equal one. The 18 corrected secondary discoveries all involve all-event set distance versus set-only gain, where an unchanged set mechanically creates a no-op. They are not treated as independent predictive evidence.

Figure A7 displays the complete changed-set metric family. A separately specified pair/boundary predictor has AUROC 0.502, 0.510, 0.499, and 0.497 in the same setting order.

A.7 LOCAL AND TASK-LEVEL SOURCE-ROUTE REPLAY

The token-local experiment replaces one changed route at one layer, keeping that event’s merged input and expert parameters fixed while subsequent computation responds. Positive ΔNLL means native NLL minus replay NLL. All four intervals in Table A4 are negative.

For task evaluation, every answer choice is teacher-forced. The source schedule replaces routing at all non-padding question-choice tokens and sparse layers, but only answer tokens enter the length-normalized score in Eq. 12. Science uses ARC-Challenge; mathematics and computer science each

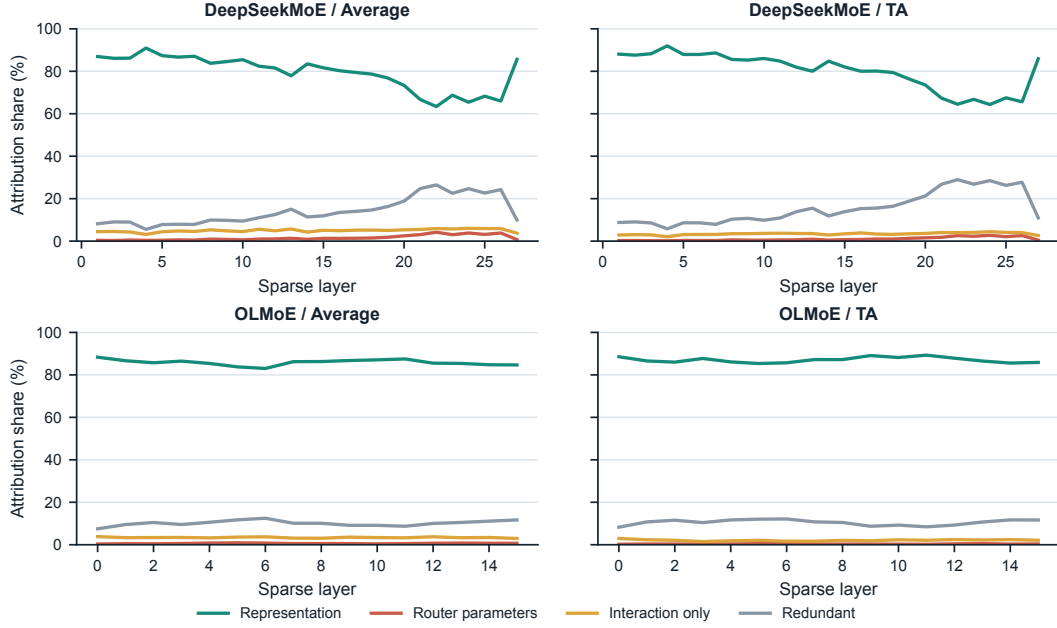


Figure A4: **Attribution across sparse layers.** Four-state shares among changed routes for DeepSeekMoE/OLMoE under Average and TA. Categories are defined in Appendix A.1.

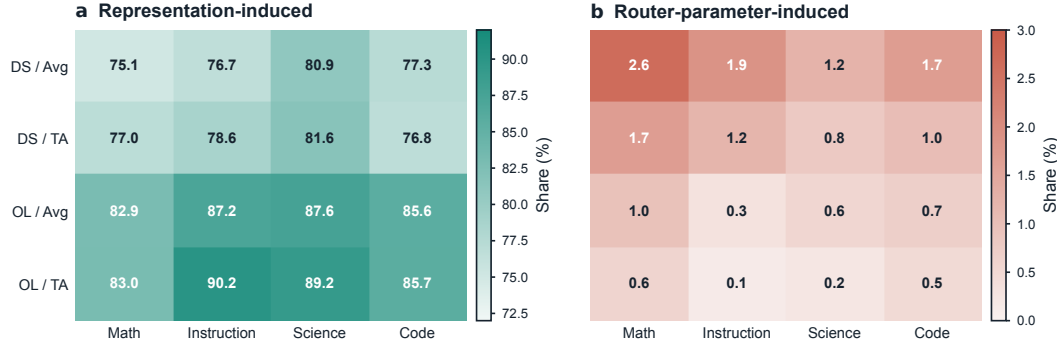


Figure A5: **Attribution by domain.** Representation-induced and router-parameter-induced shares use separate color scales; each cell reports its percentage.

comprise four MMLU subjects with 64 items per subject. Outcome-independent SHA-256 ranking fixes the items, and each domain uses its matching specialist. All 12 margin and accuracy intervals in Table A5 include zero.

A.8 QWEN3-MOE MEASUREMENTS

Qwen3-30B-A3B-Base is evaluated under Average and TA with coefficient 0.5625. Its 256 trajectories per parent retain an earlier 64-prompt sample and add 192 greedy continuations capped at 96 new tokens. Additional selection rules were fixed before collecting the added outcomes, but this extension followed earlier diagnostics; it does not change the original 16-test primary family.

Eight hash-selected events per prompt are drawn from layers 43–47 without duplicating short trajectories. Average yields 2,045 events and TA 2,048; the changed-set subsets contain 725 events from 235 prompts and 858 from 240 prompts. Table A6 retains all three available metrics. Its 4,000-draw prompt-bootstrap intervals are pointwise: all correlation intervals include zero and all AUROC intervals include 0.5.

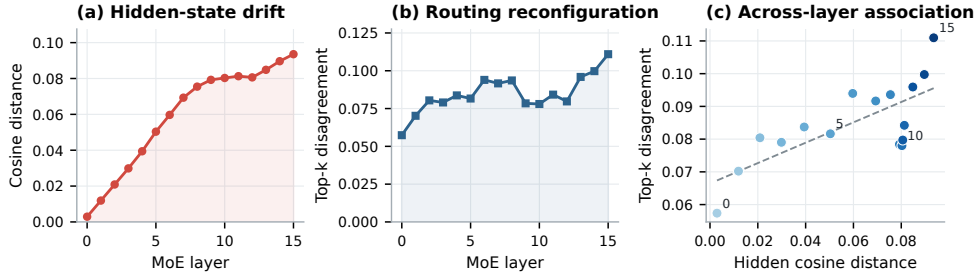


Figure A6: **Representation and routing across OLMoE depth.** Original-space hidden-state cosine distance and Top- k disagreement co-vary across 16 layers. This association is distinct from the crossed intervention.

Table A3: Primary AUROC comparisons with 95% prompt-bootstrap intervals. Route/state denote source-routing/source-hidden-state gains; 0.5 is chance. None of the 16 comparisons survives Holm correction.

Architecture / Parent	Route / Top- k	Route / JS	State / Top- k	State / JS
DeepSeekMoE / Average	.490 [.463, .517]	.510 [.472, .548]	.502 [.476, .528]	.509 [.472, .547]
DeepSeekMoE / Task Arithmetic	.491 [.465, .517]	.510 [.478, .542]	.503 [.477, .529]	.511 [.477, .547]
OLMoE / Average	.509 [.483, .533]	.519 [.482, .556]	.515 [.489, .541]	.520 [.481, .559]
OLMoE / Task Arithmetic	.525 [.502, .548]	.483 [.450, .515]	.525 [.502, .549]	.487 [.454, .519]

The all-layer attribution uses every continuation-predicting position and all 48 sparse layers, selecting eight of 128 experts. Average and TA contribute 1,081,728 and 1,105,824 records, respectively. Attribution ratios are recomputed in 10,000 prompt resamples, rather than averaging prompt percentages (Table A7).

Task replay uses the same 64 ARC-Challenge items and 256 answer choices for both parents, with interventions at all non-padding tokens and all sparse layers. Each parent loses one correct answer and gains none; 10,000 paired item resamples give the intervals below. No mathematics or computer-science replay is included in this Qwen3 sample.

A.9 LOCAL REFERENCE QUALITY

Each event has 7–12 unique candidate routes: native, source, the two crossed states in Eq. 9, a frozen linear-calibration route, and deterministic one-expert swaps. Candidates use identical merged inputs and experts. “Source wins” is a strict NLL comparison; “Merged best” admits ties within 10^{-6} . Table A9 shows limited native optimality without a reliable source advantage. This is a bounded candidate comparison, not an exhaustive route search.

A.10 MIXTURE OUTPUTS AND MATCHED CONTROLS

One event per prompt is hash-selected from the final-five-layer pool before classifying route changes. Within each setting, expert-pair, observed-mixture, and one-swap measurements use the same changed events (Table A10). Expert-pair similarity is the maximum entering-leaving output cosine. Mixtures use identical merged inputs and expert parameters; DeepSeekMoE’s fixed shared-expert branch is excluded. The 727 event-wise mixture-minus-expert gaps are all positive, with range $[0.078, 1.417]$. Gaps can exceed one because an expert-output cosine can be negative; no tail events are discarded.

Unchanged-route references are the complementary selected events, not additional interventions on the changed-event sample. Their reported mixture cosines are retained in Table A11.

One-expert swaps. The lowest-logit selected expert is replaced from the parent/source Top- $(k+4)$ union. Parent/source ranks order at most seven alternatives, independently of their outputs. Parent probabilities on the candidate set are scaled to native routed mass and stored in BF16. We aver-

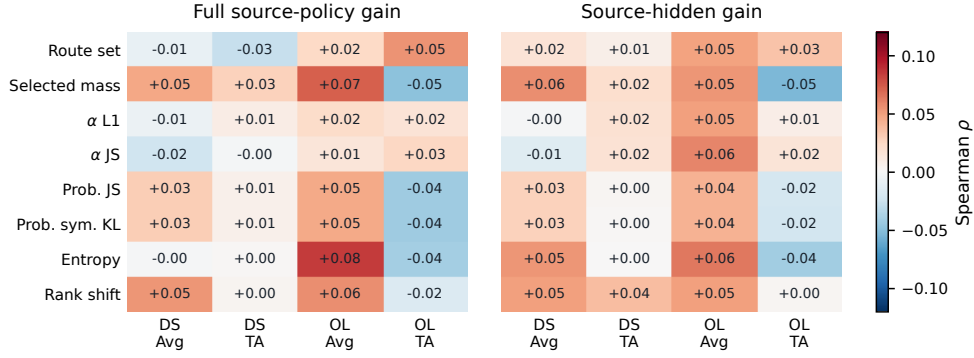


Figure A7: **Complete changed-set predictor family.** Spearman associations with source-policy gains (left) and source-hidden-state gains (right). Primary predictors are set distance and full-distribution JS.

Table A4: Token-local source-route replay. Positive native-minus-replay NLL favors replay. Intervals resample prompts.

Architecture / Parent	Δ NLL [95% CI]	Events / prompts
DeepSeekMoE / Average	-0.0339 [-0.0675, -0.0062]	80 / 71
DeepSeekMoE / Task Arithmetic	-0.0326 [-0.0682, -0.0038]	80 / 74
OLMoE / Average	-0.1623 [-0.4078, -0.0026]	80 / 66
OLMoE / Task Arithmetic	-0.1076 [-0.1947, -0.0400]	80 / 68

age within events before averaging prompts. Qwen3 excludes native/crossed sets; the DeepSeekMoE/OLMoE list additionally excludes its LC candidate. This reference-family difference does not enter the matched-random contrast.

Matched random routes. Each event has 32 alternatives preserving cardinality, the shared subset with the merged route, the exact BF16 source-weight multiset, routed mass, and concentration. Hashes determine expert alternatives and weight-slot assignments without inspecting outputs. The paired contrast subtracts mean random-route cosine from observed cosine within each event, followed by 10,000 prompt resamples. DeepSeekMoE/OLMoE random controls use a separate explicit-dispatch run; their absolute observed means must not be replaced by those in Table A10. Qwen3 uses one run for both comparisons. Table A12 preserves these within-run quantities. Cosine measures direction, not output magnitude or task recovery.

A.11 NATIVE ROUTING AND FIGURE CONVENTIONS

OLMoE uses full softmax followed by Top-8, without additional renormalization unless required by its configuration. DeepSeekMoE uses Top-6 and preserves the shared branch through routed-output delta replacement. Qwen3 uses FP32 softmax, sorted Top-8, native renormalization, and BF16 dispatch weights. The four DeepSeekMoE/OLMoE parents and eight specialists pass native-route round trips with zero NLL and output-recomputation error; the 12 corresponding task cells also reproduce native routing. An OLMoE TA boundary audit records one tied one-swap discrepancy, also present in source self-replay.

For Qwen3, unsorted replay failed the original tolerance and was replaced by native sorted dispatch before the expanded diagnostic outcomes were inspected. Final functional and task replays report zero reconstruction error under the unchanged 2×10^{-3} tolerance. Mixture comparisons capture selected experts from their original native dispatch and evaluate unselected experts at the same saved input, using one output table for all routes. Raw batched recomputation differs by up to 0.002727 in relative norm; captured native outputs reconstruct exactly.

Figure 4 classifies Qwen3 events using native GPU selections. Exact BF16 boundary ties cause 16/17 CPU–GPU discrepancies and 9/7 changed-set reclassifications for Average/TA. In contrast,

Table A5: Source-route task effects on 256 items per row. Margin and accuracy differences favor replay when positive; accuracy is a fraction. Intervals use paired item bootstrap. DS/OL denote DeepSeekMoE/OLMoE.

Model / Parent	Domain	Margin Δ [95% CI]	Accuracy Δ [95% CI]
DS / Average	Science	−0.0052 [−0.0182, +0.0079]	+0.0000 [−0.0156, +0.0156]
	Mathematics	−0.0122 [−0.0259, +0.0020]	−0.0234 [−0.0547, +0.0078]
	Computer science	−0.0005 [−0.0146, +0.0132]	+0.0234 [−0.0117, +0.0586]
DS / Task Arithmetic	Science	+0.0021 [−0.0118, +0.0170]	+0.0078 [−0.0078, +0.0234]
	Mathematics	−0.0107 [−0.0259, +0.0044]	−0.0273 [−0.0664, +0.0117]
	Computer science	−0.0005 [−0.0161, +0.0146]	+0.0312 [−0.0039, +0.0664]
OL / Average	Science	+0.0025 [−0.0261, +0.0312]	+0.0117 [−0.0156, +0.0391]
	Mathematics	+0.0046 [−0.0342, +0.0442]	−0.0117 [−0.0547, +0.0273]
	Computer science	+0.0200 [−0.0266, +0.0652]	+0.0195 [−0.0078, +0.0469]
OL / Task Arithmetic	Science	−0.0062 [−0.0418, +0.0285]	+0.0117 [−0.0117, +0.0352]
	Mathematics	+0.0525 [−0.0063, +0.1130]	+0.0000 [−0.0430, +0.0430]
	Computer science	+0.0034 [−0.0605, +0.0664]	−0.0117 [−0.0469, +0.0195]

Table A6: Qwen3 structural diagnostics on changed-set events. Intervals are pointwise 95% prompt-bootstrap intervals. Selected-weight JS aligns weights on the union of selected experts.

Parent	Distance	Spearman ρ [95% CI]	AUROC [95% CI]
Average	Full-routing JS	−.036 [−.112, .038]	.474 [.431, .518]
	Top- k set	.015 [−.061, .092]	.506 [.473, .540]
	Selected-weight JS	−.022 [−.099, .058]	.482 [.437, .530]
Task Arithmetic	Full-routing JS	.028 [−.045, .103]	.514 [.475, .553]
	Top- k set	−.009 [−.081, .058]	.500 [.469, .530]
	Selected-weight JS	.037 [−.039, .108]	.522 [.483, .561]

Figure 3 retains its historical CPU route labels. Earlier and added events retain their respective source-route caches. These conventions, as well as the sampling frames, must be retained when comparing counts across figures.

Figure 2(a) reuses 64 outcome-independent event identities across its three replacement rows within each setting; columns are not matched across architectures. Its color scale is linear on $[0, 1]$. The displayed set distance is $1 - |S_{\text{src}} \cap S_{\text{cond}}|/k$; Qwen3’s functional set-distance analysis uses one minus Jaccard similarity. Figure 3 retains all 5,762 finite changed-set observations without jitter, added epsilon, or clipping. JS uses a log axis and gains a symmetric-log axis, with linear region $[-0.002, 0.002]$. Ranks and AUROC are computed from raw values, with larger JS predicting positive gain; plotted transformations do not change the tests.

A.12 VARIATION OUTSIDE MERGING

Source-base set distances exceed source-merged distances, while different specialist routers on the same source state still disagree (Table A13). Routing variation therefore also occurs outside merging. These controls qualify the setting-specific interpretation without changing the local test.

Table A7: Qwen3 routing attribution (%; 95% prompt-bootstrap intervals). The first row includes all events; remaining rows condition on a source-merged set change.

Quantity	Average	Task Arithmetic
Overall flip rate	26.46 [25.01, 27.96]	33.20 [31.48, 34.94]
Representation-induced	96.87 [96.75, 96.98]	96.90 [96.80, 97.01]
Router-parameter-induced	0.034 [0.027, 0.041]	0.040 [0.033, 0.047]
Interaction-required	0.929 [0.873, 0.989]	0.774 [0.728, 0.823]
Redundantly sufficient	2.171 [2.086, 2.259]	2.282 [2.207, 2.357]

Table A8: Qwen3 source-route replay on 64 ARC-Challenge items. Positive changes favor replay; accuracy changes are fractions. Intervals use paired item bootstrap.

Parent	Margin Δ [95% CI]	Accuracy Δ [95% CI]
Average	+0.00880 [−.01196, +.02822]	−.015625 [−.046875, 0]
Task Arithmetic	+0.00180 [−.01685, +.01968]	−.015625 [−.046875, 0]

 Table A9: NLL reference quality within the fixed local candidate set. Source wins are strict; merged-best counts admit ties within 10^{-6} . Intervals resample prompts.

Model / Parent	Source wins [95% CI]	Ties	Merged best [95% CI]	Candidates, mean [range]
DS / Average	49.6 [43.4, 55.9]%	3.1%	10.9 [7.0, 14.8]%	9.34 [7, 12]
DS / Task Arithmetic	48.8 [43.0, 55.1]%	0.4%	12.9 [9.0, 17.2]%	9.69 [7, 12]
OL / Average	50.0 [43.8, 56.2]%	0.8%	13.3 [9.4, 17.6]%	9.31 [7, 12]
OL / Task Arithmetic	45.7 [39.8, 52.0]%	2.0%	9.8 [6.2, 13.7]%	9.55 [7, 12]

Table A10: Directional mixture comparisons on matched changed-route events. Values are mean cosines with 95% prompt-bootstrap intervals. DS/OL/QW denote DeepSeekMoE/OLMoE/Qwen3-MoE.

Model / Parent	Events	Best expert pair	Source-route mixture	One-swap mixture
DS / Average	115	0.050 [0.035, 0.067]	0.913 [0.887, 0.936]	0.951 [0.936, 0.964]
DS / Task Arithmetic	149	0.043 [0.031, 0.056]	0.896 [0.867, 0.923]	0.962 [0.950, 0.972]
OL / Average	112	0.074 [0.050, 0.101]	0.976 [0.969, 0.982]	0.990 [0.988, 0.992]
OL / Task Arithmetic	142	0.084 [0.062, 0.108]	0.965 [0.956, 0.972]	0.987 [0.984, 0.989]
QW / Average	91	0.041 [0.019, 0.065]	0.953 [0.936, 0.967]	0.980 [0.974, 0.984]
QW / Task Arithmetic	118	0.096 [0.066, 0.132]	0.957 [0.940, 0.970]	0.975 [0.964, 0.983]

Table A11: Mixture cosine on unchanged-route reference events. These events are complementary to, not matched replicates of, the changed-route sample. Values are means with 95% prompt-bootstrap intervals.

Architecture	Average	Task Arithmetic
DeepSeekMoE	0.997 [0.994, 0.999]	0.998 [0.997, 0.999]
OLMoE	0.997 [0.996, 0.998]	0.996 [0.995, 0.997]
Qwen3-MoE	1.000 [0.999, 1.000]	1.000 [0.999, 1.000]

Table A12: Observed source routing versus 32 matched random alternatives per event. Intervals apply to the paired observed-minus-random cosine, not separately to the absolute means.

Architecture / Parent	Observed source	Matched random	Difference [95% CI]
DeepSeekMoE / Average	0.902	0.769	+0.133 [+0.109, +0.158]
DeepSeekMoE / Task Arithmetic	0.895	0.748	+0.147 [+0.127, +0.168]
OLMoE / Average	0.976	0.855	+0.122 [+0.102, +0.143]
OLMoE / Task Arithmetic	0.966	0.854	+0.112 [+0.095, +0.129]
Qwen3-MoE / Average	0.953	0.840	+0.113 [+0.099, +0.128]
Qwen3-MoE / Task Arithmetic	0.957	0.841	+0.116 [+0.102, +0.131]

Table A13: Set-distance controls outside merging, with 95% prompt-bootstrap intervals.

Contrast	DeepSeekMoE	OLMoE
Source–Average	0.149 [0.138, 0.160]	0.114 [0.106, 0.122]
Source–TA	0.209 [0.195, 0.224]	0.159 [0.150, 0.168]
Base–source	0.271 [0.255, 0.287]	0.215 [0.204, 0.226]
Source–source router only	0.058 [0.053, 0.063]	0.022 [0.019, 0.024]

B SRR ALGORITHM AND FITTING OBJECTIVE

Unlike model fusion (Gu et al., 2026b; Wang et al., 2026f), SRR changes only selected rows of the merged parent’s router; experts and all other parameters remain fixed. SRR is a source-informed *candidate construction*, not a test that source routing is beneficial. It changes only selected rows of the merged parent’s router; experts and all other parameters remain fixed. The same parent-generated continuations are scored by the base, each domain specialist, and the parent, each using its own hidden states (Section 5).

SRR construction (pseudocode).

Input	Base B , specialists $\{S_d\}$, merged parent M , domain-labeled prompts, and fixed SRR settings.
	1 Generate continuations with M for fixed construction-train and validation prompt splits. Cache aligned token scores and router traces from B, S_d, and M. 2 In each of the final five sparse layers, compute the weighted source–base prompt profiles (Eq. 3). Require sign consistency ≥ 0.75 and activity ≥ 0.002 in both splits; select two disjoint expert pairs per domain. 3 For each selected pair j, form the clipped source–parent target $\tilde{r}_{tj}$, reliability weight ρ_{tj}, and parent input h_t^M on construction-train tokens. Require positive-weight support. 4 Solve the pairwise weighted ridge objective below in FP64; stop without a candidate if a pair fails to converge. 5 Apply all equal-and-opposite pair updates jointly to the FP32 parent router, then verify the changed-layer scope after BF16 conversion.
Output	SRR candidate checkpoint and fit audit; no benchmark score enters selection or fitting.

Complete fitting objective. For pair $j = (a_j, b_j)$ assigned to domain d_j , let $S_t = S_{d_t}$, $m_{tj}^A = z_{t,a_j}^A - z_{t,b_j}^A$, and $\delta_t = n_t^M - n_t^{S_t}$. The target and weight are

$$\begin{aligned}
 r_{tj} &= m_{tj}^{S_t} - m_{tj}^M, \\
 \tilde{r}_{tj} &= \text{sgn}(r_{tj}) \min\{|r_{tj}|, |m_{tj}^{S_t} - m_{tj}^B|\}, \\
 \rho_{tj} &= \mathbf{1}[d_t = d_j] \mathbf{1}[\delta_t > 0] \sigma(\delta_t / \tau).
 \end{aligned} \tag{10}$$

For N pooled construction-train response tokens, including those with zero weight, the independent fit for each pair is

$$\begin{aligned}
 v_j^* &= \arg \min_v \frac{1}{2} \sum_{t=1}^N \rho_{tj} ((h_t^M)^\top v - \tilde{r}_{tj})^2 + \frac{\lambda_j}{2} \|v\|_2^2, \\
 \lambda_j &= \gamma \max \left\{ \frac{1}{N} \sum_{t=1}^N \rho_{tj} \|h_t^M\|_2^2, 10^{-12} \right\}.
 \end{aligned} \tag{11}$$

Nonmatching domains receive *zero weight*, not a zero-change target. The fitted vector changes rows a_j and b_j by $+\eta v_j^*/2$ and $-\eta v_j^*/2$, respectively (Eq. 6).

Fit and execution. We use $\tau = 0.5$, $\gamma = 10^{-3}$, and $\eta = 0.0625$ throughout. Diagonal-preconditioned conjugate gradients solves each ridge system in FP64, checking the true relative residual against 10^{-6} within at most 4096 iterations. All fits use unchanged parent traces and precede joint writeback; there is no gradient-training epoch or trajectory refresh. The validation split screens pairs and audits the fixed-input logit change, but does not tune the update scale. Pair scarcity, absent positive support, solver failure, or an incorrect BF16 changed-layer scope aborts construction rather than triggering a benchmark-dependent fallback. The objective matches selected router-logit gaps, not task loss; successful fitting alone does not imply downstream recovery (Appendix D.4). Further selection and solver details appear in Appendix D.

C TASK-GROUNDED INTERVENTION PROTOCOLS

This appendix reports the scoring convention, controlled corruption, and natural LC replay used in Section 4. Source-specialist task replay is reported separately in Appendices A.7 and A.8. Each effect is paired within its own model, item set, and routing reference.

Table C14: Original OLMoE replay controls. Accuracy changes are percentage points; margins retain score units. Brackets are 95% paired item-bootstrap intervals. The injected and LC comparisons are distinct experiments.

Parent	Accuracy change [interval]	Margin change [interval]
<i>Injected corruption → clean-native routing replay</i>		
Average	+28.125 [+20.313, +35.547]	+0.7677 [+0.5411, +0.9934]
Task Arithmetic	+28.125 [+21.094, +35.156]	+0.5588 [+0.3675, +0.7549]
<i>Native merged routing → frozen LC routing replay</i>		
Average	−0.391 [−1.172, 0.000]	−0.0003 [−0.0037, +0.0032]
Task Arithmetic	+0.391 [0.000, +1.172]	−0.0026 [−0.0066, +0.0008]

C.1 TASK SCORING AND PREDICTION TRANSITIONS

For choice c with answer tokens y_c , define

$$s_c = -\frac{1}{|y_c|} \sum_t \text{NLL}(y_c, t), \quad m = s_{c^*} - \max_{c \neq c^*} s_c. \quad (12)$$

The prediction maximizes s_c ; only answer tokens contribute to scoring. Task labels evaluate a specified routing intervention, not a separately chosen reference for each correct answer. Margin is a continuous score readout, distinct from accuracy. For N paired items,

$$\Delta \text{Acc} = \frac{n_{\text{fix}} - n_{\text{break}}}{N}, \quad (13)$$

where the counts denote wrong-to-correct and correct-to-wrong transitions. Multiplying by 100 converts accuracy differences to percentage points. Intervals use 10,000 paired item-bootstrap draws.

C.2 FULL-LAYER AND GRADED CORRUPTION CONTROLS

The original control permutes OLMoE expert logits at layers 0–15 and replays the same parent’s known clean schedule. The fixed seeds are 2026091701 (Average) and 2026091702 (TA), with 256 items per parent. Both comparisons increase accuracy from 22.65625% to 50.78125%, corresponding to 72 net corrections, but not necessarily identical prediction transitions. Table C14 preserves accuracy and margin effects. The original injected-corruption comparisons correct 94 and 90 answers and break 22 and 18 under Average and TA, respectively; the separate natural LC comparisons correct/break 0/1 and 1/0.

The graded experiment uses the same archived parents and 256 matched ARC items, but a separate randomization. Seeds 17017 and 29029 fix nested layer orders; the first $k = 1, 4, 16$ layers receive a logit permutation. Table C15 retains every condition, pointwise paired interval, and the Holm decision across twelve tests. All eight four- and sixteen-layer recoveries pass the specified correction; none of the one-layer tests does. Seeds are fixed corruption realizations, not independent trained models. Replay ranges from 50.39% to 51.56% around the 50.78% clean reference and does not establish exact reconstruction. These controls concern the imposed losses, not learned SRR repair or a universal sensitivity threshold.

C.3 NATURAL LC REPLAY AND NUMERICAL VALIDITY

A separate OLMoE cohort replays schedules from frozen LC-MERGE-RC router checkpoints on 256 ARC items per parent. All non-routing parameters remain those of the native parent. The archive name *LC* distinguishes this fixed-sequence replay from end-to-end SRR evaluation. Accuracy changes by -0.390625 pp for Average and $+0.390625$ pp for TA: one changed-correctness item per parent in opposite directions. Both accuracy intervals include zero; both mean margin changes are negative with intervals crossing zero. The natural and injected cohorts are not pooled.

The original positive-control bootstrap estimates and transition counts were reproduced from the archived item records and recorded seeds, without model inference. The OLMoE LC identity replay

Table C15: Graded OLMoE corruption on 256 items per parent. Accuracy is in %; recovery and its paired 95% interval are in pp. Clean accuracy is 50.78%. Holm decisions cover all twelve parent–seed–scope comparisons.

Parent	Permutation seed	k	Corrupted	Replay	Recovery [95% CI]	Holm $p < .05$
Average	17017	1	53.12	50.78	−2.34 [−6.25, +1.56]	No
		4	38.28	50.78	+12.50 [+7.03, +18.36]	Yes
		16	17.97	50.78	+32.81 [+25.39, +40.23]	Yes
	29029	1	49.22	50.39	+1.17 [−3.52, +5.86]	No
		4	37.89	50.78	+12.89 [+6.64, +18.76]	Yes
		16	23.44	50.78	+27.34 [+19.92, +34.77]	Yes
TA	17017	1	52.73	50.78	−1.95 [−5.86, +1.95]	No
		4	36.33	51.17	+14.84 [+9.38, +20.70]	Yes
		16	19.92	51.56	+31.64 [+23.83, +39.45]	Yes
	29029	1	48.83	51.17	+2.34 [−1.95, +6.64]	No
		4	36.72	51.17	+14.45 [+8.59, +20.31]	Yes
		16	24.61	51.17	+26.56 [+18.75, +33.98]	Yes

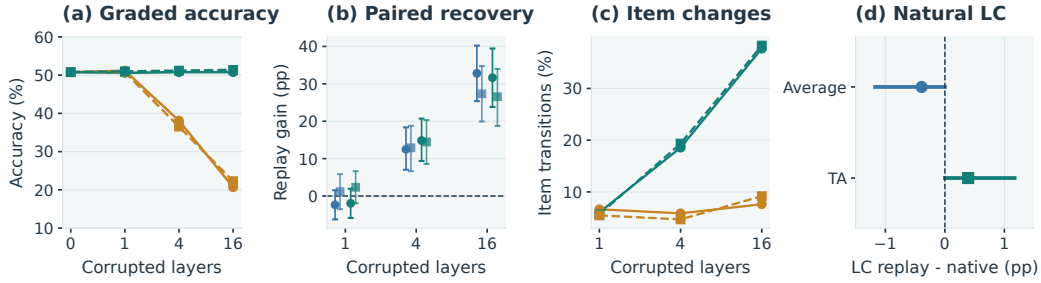


Figure C8: **Graded routing interventions.** (a) Clean, corrupted, and replay accuracy, averaging two fixed permutations per parent. (b) All twelve paired recovery effects with 95% item-bootstrap intervals. (c) Corrections and regressions, averaged within parent and scope. (d) Independent natural LC effects. Seeds are not model replicates.

reports zero native-recomputation relative error under tolerance 2×10^{-3} . This DeepSeekMoE check concerns only the separate LC schedule-replay experiment, not the end-to-end SRR evaluation in Table 2.

D SRR IMPLEMENTATION AND PROXY CHECKS

Section 5 defines the profile, target, regression, and writeback. Here we specify the remaining selection and numerical details, then retain two checks distinguishing proxy fit from task recovery. The fixed values are collected in Table D16; none is selected using benchmark outcomes.

D.1 DETERMINISTIC EXPERT-PAIR SELECTION

The implementation forms the profile from centered log-softmax differences. Centering removes the common offset, giving $\text{center}(\log \text{softmax}(z^S) - \log \text{softmax}(z^B)) = \text{center}(z^S - z^B)$ in exact arithmetic. Unlike fitting weights, profile weights ω_t are not thresholded at positive advantage. Expert activity is

$$a_{ie} = \frac{\sum_{t \in i} \omega_t \max\{\text{softmax}(z_t^S)_e, \text{softmax}(z_t^B)_e\}}{\max\{\sum_{t \in i} \omega_t, 10^{-8}\}}. \quad (14)$$

The same denominator floor applies to the prompt profile in Eq. 3.

For each domain and split $s \in \{\text{tr}, \text{val}\}$, let μ_e^s and A_e^s be prompt means of the profile and activity, and $f_{e,+}^s, f_{e,-}^s$ the fractions with positive and negative profile. An eligible positive expert satisfies $\mu_e^s > 0$, $f_{e,+}^s \geq \kappa$, and $A_e^s \geq a_{\min}$ in both splits; negative eligibility is symmetric. Pairs (a, b) are ranked by

$$q_{ab} = \min_s (\mu_a^s - \mu_b^s) \min_s \{f_{a,+}^s, f_{b,-}^s\} \sqrt{\min_s \{A_a^s, A_b^s\}}. \quad (15)$$

Expert indices break ties. In fixed domain order, selection takes the highest-ranked unused pair, without reusing endpoints within a layer. Construction stops if a domain cannot meet its quota or a selected pair has no positive source-over-parent training support; there is no benchmark-dependent fallback.

D.2 NUMERICAL SOLVE AND WRITEBACK

Let H stack training inputs and $D_j = \text{diag}(\rho_{\cdot,j})$. The regression in Eq. 5 solves

$$(H^\top D_j H + \lambda_j I) v_j = H^\top D_j \tilde{r}_j. \quad (16)$$

Its data term is a weighted sum. The N in the ridge scale counts all pooled training response tokens, including zero-weight tokens for that pair. Nonmatching-domain tokens have zero weight, not a positively weighted zero-change preservation target.

Batched FP64 preconditioned conjugate gradients maintains a separate convergence state per pair. Matrix products use $H^\top [\rho_{\cdot,j} \odot (Hv)] + \lambda_j v$ and the diagonal $\sum_t \rho_{tj} (h_t^M \odot h_t^M) + \lambda_j \mathbf{1}$ as preconditioner. The true system residual is recomputed, and every selected pair must converge before writeback. All fits use unchanged parent traces; updates are then applied jointly. FP32 router weights are saved and changed-layer scope is checked after BF16 conversion. The covariance-based logit-change budget is diagnostic, not an enforced projection. Sparse row updates do not guarantee unchanged probabilities or expert choices elsewhere.

D.3 HELD-OUT WEIGHTING ABLATION

A separate DeepSeekMoE/OLMoE Average-TA experiment varies the residual-fitting weights. Table D17 reports all-held-out MSE, not task performance. All-uniform has the lowest reported MSE

Table D16: Fixed SRR settings. Pair quotas apply to each domain at each eligible layer.

Quantity	Value	Quantity	Value
Likelihood temperature τ	0.5	Relative ridge γ	10^{-3}
Sign consistency κ	0.75	Update scale η	0.0625
Minimum activity $a_{\min}$	0.002	Profile/activity floor	10^{-8}
Pairs per domain/layer	2	Ridge energy floor	10^{-12}
Eligible router layers	Final 5	PCG iteration limit	4096
True-residual tolerance	10^{-6}		

in all four settings; weighted-positive beats shuffled-positive in one. These summaries lack paired prompt uncertainty and complete support-normalization details, so they do not establish the task-level superiority of any weighting rule. This experiment is separate from the local intervention sample in Appendix E.

Table D17: Held-out residual-fitting MSE in a separate weighting ablation; lower is better. These point estimates are not task scores and have no paired uncertainty. DS/OL denote DeepSeek-MoE/OLMoE.

Cell	Weighted positive	Uniform positive	All uniform	Shuffled positive
DS / Average	.01774	.01682	.01624	.01761
DS / TA	.05171	.05065	.04973	.05406
OL / Average	.06645	.05756	.05471	.06211
OL / TA	.11473	.10365	.09864	.10738

D.4 A COUNTEREXAMPLE TO TASK IMPROVEMENT FROM PROXY FITTING

Consider a hypothetical normalized Top-1 layer with $h = 1$, $z^M = z^B = (0, \epsilon)$ and $z^S = (2\epsilon, 0)$, where $\epsilon > 0$. Parent experts b and a assign the correct token probabilities 0.8 and 0.1; source expert a assigns 0.9, and the base may assign 0.6. The source therefore has positive likelihood advantage, while the selected pair residual and its clipping bound both equal 3ϵ . With one fitting token, $\gamma = 1$ and an inactive numerical floor, $\lambda = \rho > 0$ and $v^* = 3\epsilon/2$. At $\eta = 1$, the new parent logits are $(3\epsilon/4, \epsilon/4)$: the fitting error falls to one quarter, but routing switches to the worse merged expert and NLL rises by $\log 8$. This construction is not an experiment or a claim about the frozen hyperparameters. It shows why the proxy alone supplies no general task-recovery guarantee when source and merged expert functions differ.

E LOCAL UTILITY OF SOURCE-INFORMED CORRECTIONS

E.1 INDEPENDENT SAMPLE AND PAIRED CONTRASTS

Related on-policy distillation work cautions that token-level disagreement need not be learnable (Wang et al., 2026e); here we test the distinct question of whether source-likelihood advantage identifies beneficial router corrections. The local diagnostic covers the four OLMoE/Qwen3-MoE Average-TA settings, using 256 independent prompts per candidate and one hash-selected token-layer-pair event per prompt. Positive source advantage occurs in 101, 122, 117, and 131 events, respectively: 471 of 1,024 events. Construction validation is used for pair selection, not as this independent diagnostic sample.

Each event compares native routing with equal-magnitude positive and opposite logit perturbations in the raw source-informed and fitted SRR directions. The U and D contrasts follow Eq. 7; $D > 0$ does not by itself establish improvement over native. Intervals use 10,000 prompt-bootstrap draws with the fixed analysis seed. The tables retain positive-support, non-positive-support, and all-event results separately. No local NLL result is converted into task accuracy.

E.2 MATCHED SUPPORT AND DIRECTION PRESERVATION

Positive events are matched with replacement to non-positive events from the same architecture, merge, role, layer, and expert pair, choosing the nearest native NLL. The 440 matched pairs yield source-direction positive-minus-matched $U = -0.000341$ nats/token (95% interval $[-0.003323, 0.002631]$) and $D = -0.000997$ ($[-0.005653, 0.003868]$). These contrasts differ from positive-support means and do not establish enrichment.

A fixed-effect regression of continuous sigmoid weight beyond positive support has coefficient $+0.0720$ (CI $[-0.0279, 0.1985]$). Source and fitted directions agree in sign on 83.4% of supported events (CI $[80.0, 86.6]\%$), but agreement does not establish local benefit. Only two of four settings passed the prespecified source-direction sign screen; the contingent original scale-up was therefore

Table E18: **Local intervention gains by support group.** Entries are estimates and 95% prompt-bootstrap intervals in 10^{-3} nats/token. U compares the proposed direction with native routing; D compares it with the opposite direction. Support groups are reported separately, not pooled as independent samples.

Setting	n	Source U	Source D	Fitted U	Fitted D
Positive support					
OL / Average	101	+1.38 [−2.18, +5.14]	+2.77 [−4.52, +10.60]	+2.08 [−1.40, +5.87]	+4.16 [−3.27, +12.07]
OL / TA	122	−0.88 [−6.01, +4.52]	−4.59 [−14.68, +7.04]	+0.69 [−4.29, +6.05]	−1.46 [−11.29, +10.24]
QW / Average	117	−0.56 [−2.21, +0.95]	−1.44 [−3.31, −0.08]	−0.56 [−2.19, +0.94]	−1.44 [−3.26, −0.06]
QW / TA	131	+0.49 [−3.51, +4.58]	+1.44 [−3.95, +7.34]	+0.49 [−3.48, +4.71]	+1.44 [−3.85, +7.45]
Pooled	471	+0.11 [−1.77, +2.05]	−0.45 [−3.86, +3.37]	+0.67 [−1.24, +2.62]	+0.67 [−2.82, +4.46]
Non-positive support					
OL / Average	155	+0.26 [−4.72, +5.41]	−2.73 [−11.37, +6.03]	+0.54 [−4.47, +5.75]	−2.16 [−11.06, +6.63]
OL / TA	134	−0.15 [−4.31, +4.03]	+5.33 [−1.36, +12.94]	+2.00 [−1.99, +6.11]	+9.63 [+3.17, +17.15]
QW / Average	139	−2.30 [−4.34, −0.60]	−1.75 [−5.05, +1.11]	−1.92 [−4.08, −0.10]	−0.99 [−4.32, +1.85]
QW / TA	125	−0.19 [−4.30, +3.84]	+0.01 [−4.53, +4.47]	+0.09 [−2.40, +2.58]	+0.58 [−3.99, +5.08]
All events					
OL / Average	256	+0.70 [−2.66, +4.14]	−0.56 [−6.68, +5.65]	+1.15 [−2.39, +4.72]	+0.33 [−5.71, +6.46]
OL / TA	256	−0.50 [−3.69, +2.87]	+0.60 [−5.64, +7.35]	+1.37 [−1.75, +4.73]	+4.34 [−1.74, +11.10]
QW / Average	256	−1.50 [−2.85, −0.33]	−1.61 [−3.57, +0.09]	−1.29 [−2.68, −0.07]	−1.19 [−3.15, +0.56]
QW / TA	256	+0.16 [−2.65, +3.06]	+0.74 [−2.89, +4.43]	+0.29 [−2.10, +2.74]	+1.02 [−2.50, +4.75]

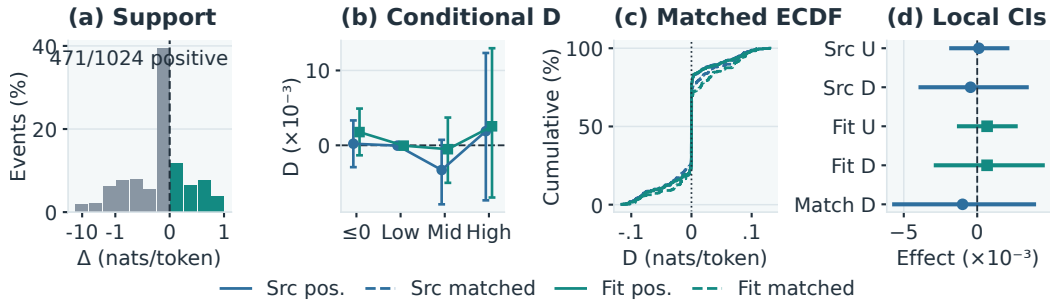


Figure E9: **Local source-supervision diagnostics.** (a) Likelihood advantage across 1,024 events (asinh-spaced axis; labels in nats/token). (b) Directional contrast D by support group. (c) Positive-support and matched-control ECDFs. (d) Local and matched effects with prompt-bootstrap intervals. These are fixed-continuation NLL measurements.

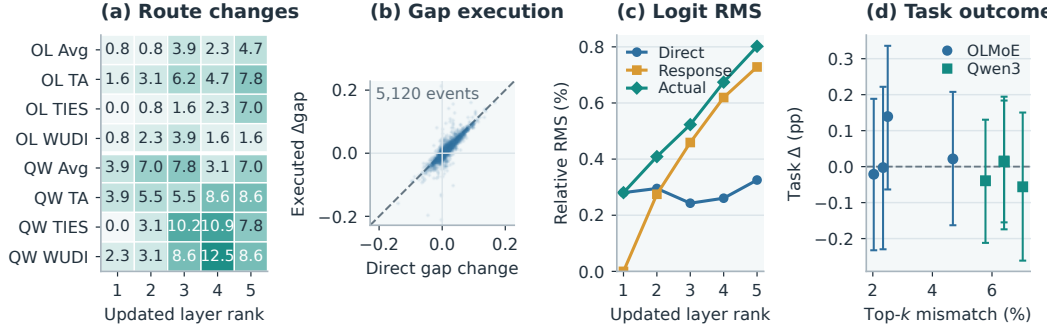


Figure F10: **Fitted and executed corrections.** (a) Native expert-set changes. (b) Direct versus executed pair gaps on 5,120 events. (c) Centered-logit RMS, normalized by each parent before equal-setting averaging. (d) Identity-matched task differences. RMS components are not additive; the task association is descriptive.

not run. The empirical distributions and support-bin analysis in Figure E9 describe these fixed local events, not additional checkpoint trials.

F EXECUTED SRR CORRECTIONS

F.1 ALIGNED TRAJECTORIES AND NUMERICAL CHECKS

The execution study uses eight OLMoE/Qwen3-MoE Parent/SRR pairs from the task evaluation. Each setting has 128 prompts (32 per domain) and one hash-selected token-pair event per prompt at each of five updated layers, totaling 5,120 events. Both models are teacher-forced on the same tokens but produce their own hidden trajectories. The direct and input-response terms follow Eq. 8; they are reconstructed in FP64 from the specified weights and states.

Inputs agree at the first updated layer. The maximum normalized algebra residual is 6.9×10^{-18} , whereas the maximum native-forward versus FP64 logit discrepancy is 0.0621. The latter is a separate measurement, not an input-response term. Algebraic agreement establishes the stated accounting identity, not bitwise native replay or the cause of task effects.

F.2 COMPLETE LAYER-WISE MEASUREMENTS

Tables F19–F22 retain every setting and updated layer. Each RMS uses 128 prompts and 10,000 prompt resamples for pointwise 95% intervals. Direct, response, and executed RMS values do not add because their signed components have a cross term. The response/direct RMS ratios in Figure 6 are ratios of reported point estimates, not additive attribution shares.

Native expert-set and dispatch-weight changes are reported separately from reference-precision logits. A nonzero correction need not change the selected set. These traces differ from the independent local-NLL sample and the task items; the cross-setting association with task effects is descriptive, not an event-wise or causal attribution.

G SUPPLEMENTARY TASK EVALUATION

Scoring and uncertainty. Each Parent/SRR pair is evaluated five times on 35,326 matched items using five-shot `lm-evaluation-harness` with a VLLM backend. Table G23 lists the task sizes and scoring fields. The aggregate equally weights the eight task percentages. For each matched item, we first average the paired SRR–Parent score difference over the five evaluation runs. Paired item bootstraps then resample item identities within each task and recompute both task differences and the aggregate. Intervals condition on fixed checkpoints; they do not capture variation across repair

Table F19: **OLMoE SRR pair-gap decomposition.** Each cell is a pair-logit-gap RMS with a pointwise 95% prompt-bootstrap interval, conditional on its fixed Parent/SRR checkpoints. RMS terms do not add.

Merge	Layer	Direct [95% CI]	Input response [95% CI]	Executed [95% CI]
Avg	11	0.0135 [0.0116, 0.0153]	0 [0, 0]	0.0135 [0.0116, 0.0153]
	12	0.013 [0.0113, 0.0147]	0.00341 [0.00272, 0.00415]	0.0135 [0.0118, 0.015]
	13	0.0138 [0.0121, 0.0155]	0.00587 [0.00463, 0.00744]	0.0148 [0.0132, 0.0164]
	14	0.0151 [0.0134, 0.0168]	0.00835 [0.00613, 0.0109]	0.0189 [0.0159, 0.0222]
	15	0.0196 [0.0165, 0.0233]	0.0077 [0.00648, 0.00896]	0.0221 [0.0187, 0.0258]
TA	11	0.0151 [0.0127, 0.0176]	0 [0, 0]	0.0151 [0.0127, 0.0176]
	12	0.0169 [0.0144, 0.0193]	0.0114 [0.00306, 0.0193]	0.0208 [0.0156, 0.027]
	13	0.016 [0.0135, 0.0184]	0.0147 [0.00621, 0.0226]	0.023 [0.0165, 0.0304]
	14	0.0147 [0.0131, 0.0163]	0.0147 [0.00697, 0.0227]	0.0191 [0.0156, 0.0234]
	15	0.0206 [0.0174, 0.0244]	0.0118 [0.00811, 0.0158]	0.0234 [0.0194, 0.0279]
TIES	11	0.0145 [0.0123, 0.0167]	0 [0, 0]	0.0145 [0.0123, 0.0167]
	12	0.0128 [0.0109, 0.0149]	0.00264 [0.00218, 0.00311]	0.0133 [0.0113, 0.0155]
	13	0.0146 [0.0122, 0.017]	0.00675 [0.00452, 0.00933]	0.0158 [0.0134, 0.0183]
	14	0.0143 [0.0122, 0.0164]	0.00779 [0.00628, 0.00937]	0.0174 [0.0144, 0.0206]
	15	0.0228 [0.0184, 0.027]	0.00877 [0.0069, 0.0111]	0.0244 [0.0198, 0.029]
WUDI	11	0.00697 [0.00622, 0.00771]	0 [0, 0]	0.00697 [0.00622, 0.00771]
	12	0.00995 [0.00825, 0.0117]	0.00244 [0.00193, 0.00298]	0.0102 [0.00856, 0.012]
	13	0.00693 [0.00603, 0.00783]	0.00713 [0.00372, 0.011]	0.00963 [0.00773, 0.012]
	14	0.00903 [0.00785, 0.0102]	0.00886 [0.00661, 0.0113]	0.0131 [0.0107, 0.0157]
	15	0.0154 [0.0136, 0.0171]	0.0133 [0.00875, 0.0181]	0.0221 [0.0175, 0.028]

Table F20: **Qwen3-MoE SRR pair-gap decomposition.** Each cell is a pair-logit-gap RMS with a pointwise 95% prompt-bootstrap interval, conditional on its fixed Parent/SRR checkpoints. RMS terms do not add.

Merge	Layer	Direct [95% CI]	Input response [95% CI]	Executed [95% CI]
Avg	43	0.0323 [0.0279, 0.0365]	0 [0, 0]	0.0323 [0.0279, 0.0365]
	44	0.0357 [0.0303, 0.0408]	0.00439 [0.00319, 0.00569]	0.0358 [0.0304, 0.0408]
	45	0.0272 [0.0213, 0.0334]	0.00511 [0.00415, 0.00604]	0.0272 [0.021, 0.0339]
	46	0.0266 [0.0233, 0.0298]	0.0233 [0.00859, 0.037]	0.035 [0.0265, 0.0452]
	47	0.0332 [0.0275, 0.0388]	0.0204 [0.0134, 0.0268]	0.0429 [0.0339, 0.0517]
TA	43	0.0353 [0.0303, 0.0402]	0 [0, 0]	0.0353 [0.0303, 0.0402]
	44	0.0407 [0.0349, 0.0465]	0.0051 [0.00366, 0.00655]	0.0412 [0.0351, 0.0474]
	45	0.0266 [0.0217, 0.0313]	0.00824 [0.00626, 0.0101]	0.0274 [0.0221, 0.0323]
	46	0.0305 [0.0267, 0.0343]	0.0151 [0.0109, 0.019]	0.0362 [0.0314, 0.0408]
	47	0.0493 [0.0419, 0.0566]	0.0185 [0.0125, 0.0254]	0.053 [0.0456, 0.0603]
TIES	43	0.0227 [0.0202, 0.0252]	0 [0, 0]	0.0227 [0.0202, 0.0252]
	44	0.0228 [0.0201, 0.0255]	0.00451 [0.00302, 0.00619]	0.0237 [0.0207, 0.0267]
	45	0.0197 [0.0168, 0.0226]	0.0066 [0.00472, 0.00855]	0.0202 [0.0176, 0.0228]
	46	0.0218 [0.0187, 0.0249]	0.0107 [0.0081, 0.0134]	0.0231 [0.0198, 0.0264]
	47	0.0317 [0.0282, 0.035]	0.0168 [0.0121, 0.0213]	0.0358 [0.0317, 0.0398]
WUDI	43	0.0308 [0.0263, 0.0351]	0 [0, 0]	0.0308 [0.0263, 0.0351]
	44	0.0314 [0.0266, 0.0359]	0.00459 [0.0032, 0.00599]	0.0319 [0.027, 0.0366]
	45	0.0283 [0.0242, 0.0325]	0.00933 [0.00688, 0.0117]	0.0294 [0.0255, 0.0334]
	46	0.029 [0.0246, 0.0332]	0.0235 [0.00905, 0.0368]	0.038 [0.0284, 0.0485]
	47	0.0271 [0.0231, 0.0313]	0.0215 [0.0149, 0.0295]	0.0376 [0.0295, 0.0467]

seeds, calibration samples, or training runs. The SRR–Parent intervals do not apply to HARC–Parent or SRR–HARC comparisons.

Supplementary selection test. A separate pilot tested 40 nonzero source, opposite, and random-direction proposals across four settings. Multiplicity-controlled choice-loss tests and task guardrails preceded evaluation on 1,536 confirmation items. All 20 family–setting decisions retained Parent, yielding zero confirmation differences by checkpoint identity. Fixed, nonselected source-scale-one updates yielded -0.016 pp (95% paired CI $[-0.179, +0.146]$). Rejection under this rule does not establish that every candidate is ineffective.

Table F21: **OLMoE SRR routing and numerical diagnostics.** One hash-selected event per prompt and updated layer. Cross is the mean centered-logit direct-response inner product. The native-FP64 difference is the largest absolute BF16-forward versus reconstructed-logit discrepancy; algebra residual is normalized. Route changes and dispatch-weight distance use native routing.

Merge	Layer	Cross	Top- <i>k</i> change (%)	Weight $\ell_1/2$	$\ \Delta h\ /\ h\ $	Native-FP64	Algebra resid.
Avg	11	0	0.781	0.00042	0	0.0185	0
	12	-1.8e-07	0.781	0.000836	0.00363	0.0211	0
	13	-6.54e-08	3.91	0.00198	0.00643	0.0247	0
	14	3.36e-07	2.34	0.00173	0.00947	0.0208	0
	15	-1.13e-07	4.69	0.00255	0.0112	0.0291	0
TA	11	0	1.56	0.000725	0	0.0207	0
	12	-1.94e-07	3.12	0.00172	0.0045	0.0212	0
	13	2.24e-07	6.25	0.00305	0.00841	0.0221	0
	14	-1.36e-07	4.69	0.00248	0.0117	0.0216	0
	15	-4.65e-07	7.81	0.00344	0.0133	0.0247	0
TIES	11	0	0	0.000249	0	0.0156	0
	12	-3.8e-08	0.781	0.000761	0.00358	0.0157	0
	13	-2.68e-07	1.56	0.00135	0.00709	0.0182	0
	14	3.47e-07	2.34	0.00195	0.00951	0.0221	0
	15	-8.7e-07	7.03	0.00345	0.012	0.0296	0
WUDI	11	0	0.781	0.00033	0	0.0201	0
	12	6.6e-08	2.34	0.00122	0.00307	0.0231	0
	13	-8.84e-08	3.91	0.0018	0.0065	0.0215	0
	14	-1.27e-07	1.56	0.00141	0.0091	0.0235	0
	15	2.39e-07	1.56	0.00172	0.011	0.0255	0

Table F22: **Qwen3-MoE SRR routing and numerical diagnostics.** One hash-selected event per prompt and updated layer. Cross is the mean centered-logit direct-response inner product. The native-FP64 difference is the largest absolute BF16-forward versus reconstructed-logit discrepancy; algebra residual is normalized. Route changes and dispatch-weight distance use native routing.

Merge	Layer	Cross	Top- <i>k</i> change (%)	Weight $\ell_1/2$	$\ \Delta h\ /\ h\ $	Native-FP64	Algebra resid.
Avg	43	0	3.91	0.00558	0	0.0451	0
	44	-7.73e-08	7.03	0.00772	0.00338	0.0526	6.87e-18
	45	-4.01e-07	7.81	0.00846	0.00635	0.0476	0
	46	-2.88e-07	3.12	0.00635	0.00871	0.0543	0
	47	-2.05e-07	7.03	0.00794	0.00917	0.0613	0
TA	43	0	3.91	0.00417	0	0.0434	0
	44	-3.53e-08	5.47	0.00691	0.00399	0.0466	0
	45	-1.68e-07	5.47	0.00779	0.00802	0.049	0
	46	3.22e-07	8.59	0.00946	0.0107	0.0527	0
	47	-7.97e-07	8.59	0.0109	0.0115	0.0611	0
TIES	43	0	0	0.00108	0	0.0453	0
	44	-1.28e-07	3.12	0.00365	0.00291	0.0498	0
	45	-8.17e-08	10.2	0.00958	0.00636	0.0452	0
	46	-3.19e-07	10.9	0.0101	0.00873	0.062	0
	47	-3.01e-07	7.81	0.00872	0.00965	0.0614	0
WUDI	43	0	2.34	0.00333	0	0.0478	0
	44	8.69e-08	3.12	0.00437	0.00339	0.0494	0
	45	-1.93e-07	8.59	0.00901	0.00671	0.0534	0
	46	3.78e-08	12.5	0.0138	0.0105	0.045	0
	47	1.3e-06	8.59	0.0102	0.0113	0.0618	0

Table G23: Evaluation items and scoring fields. Task scores receive equal weight.

Benchmark	Items	Scoring field
MMLU	14,042	acc, none
HellaSwag	10,042	acc_norm, none
ARC-Challenge	1,172	acc_norm, none
ARC-Easy	2,376	acc_norm, none
PIQA	1,838	acc_norm, none
WinoGrande	1,267	acc, none
BoolQ	3,270	acc, none
GSM8K	1,319	exact_match, flexible-extract